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**Master's Thesis
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***Predicting Last-Mile Delivery Delays Using
Machine Learning: A Comparative Model
Analysis***

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MASTER'S THESIS DECLARATION

I, the undersigned, Chiramani Saiteja, hereby declare that I have read and understood the document regarding the Intellectual Property Rules and confirm that these rules have been observed in the enclosed document.

Date: 22 May 2026

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Abstract

The rapid expansion of e-commerce has transformed global logistics operations and increased the importance of efficient last-mile delivery systems. As customer expectations continue to rise, delivery speed and reliability have become critical factors influencing customer satisfaction and competitive advantage. However, last-mile delivery remains highly vulnerable to delays caused by traffic congestion, operational inefficiencies, geographical constraints, seasonal demand fluctuations, and order-related complexities.

This study examines how machine learning techniques can be used to predict last-mile delivery delays in e-commerce logistics using large-scale operational data. The research is based on the Brazilian E-Commerce Public Dataset provided by Olist through Kaggle, which contains approximately 115,000 real-world e-commerce transactions collected between 2016 and 2018 across multiple relational tables, including orders, products, customers, sellers, and payments.

A structured data preprocessing and feature engineering pipeline was implemented in Python. This process included filtering delivered orders, handling missing values, converting datetime fields, merging relational tables, and constructing predictive features. The engineered variables included purchase time characteristics, product dimensions, freight values, payment-related variables, total order values, and product weight. The target variable, `delay_days`, was defined as the difference between the actual delivery date and the estimated delivery date.

The study developed and compared four supervised machine learning regression models: Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score. The results showed that ensemble learning methods outperformed traditional regression approaches, with the Random Forest Regressor achieving the best overall performance.

The findings highlight the practical value of machine learning for improving delivery reliability and operational efficiency in e-commerce logistics. More accurate delay prediction can help logistics providers allocate resources more effectively, improve customer communication, reduce operational costs, and enhance service quality. This study also contributes to predictive analytics in supply chain management by presenting a scalable and interpretable framework for delivery delay forecasting using real-world logistics data.

Résumé

L'expansion rapide du commerce électronique a profondément transformé les opérations logistiques mondiales et renforcé l'importance des systèmes efficaces de livraison du dernier kilomètre. À mesure que les attentes des clients augmentent, la rapidité et la fiabilité des livraisons sont devenues des facteurs essentiels de satisfaction client et d'avantage concurrentiel. Cependant, la livraison du dernier kilomètre reste fortement exposée aux retards causés par la congestion du trafic, les inefficacités opérationnelles, les contraintes géographiques, les variations saisonnières de la demande et la complexité des commandes.

Cette étude examine comment les techniques de machine learning peuvent être utilisées pour prédire les retards de livraison du dernier kilomètre dans la logistique du e-commerce à partir de données opérationnelles à grande échelle. La recherche s'appuie sur le Brazilian E-Commerce Public Dataset fourni par Olist sur Kaggle, qui contient environ 115 000 transactions réelles de commerce électronique collectées entre 2016 et 2018 à travers plusieurs tables relationnelles, notamment les commandes, les produits, les clients, les vendeurs et les paiements.

Un pipeline structuré de prétraitement des données et de feature engineering a été mis en œuvre en Python. Ce processus comprenait le filtrage des commandes livrées, le traitement des valeurs manquantes, la conversion des variables de date et d'heure, la fusion des tables relationnelles et la construction de variables prédictives. Les variables construites incluaient les caractéristiques temporelles d'achat, les dimensions des produits, les frais de transport, les variables liées au paiement, la valeur totale des commandes et le poids des produits. La variable cible, `delay_days`, a été définie comme la différence entre la date réelle de livraison et la date de livraison estimée.

L'étude a développé et comparé quatre modèles de régression supervisée en machine learning : Linear Regression, Decision Tree Regressor, Random Forest Regressor et XGBoost Regressor. La performance des modèles a été évaluée à l'aide du Mean Absolute Error (MAE), du Root Mean Square Error (RMSE) et du R^2 Score. Les résultats montrent que les méthodes d'ensemble surpassent les approches de régression traditionnelles, le Random Forest Regressor obtenant la meilleure performance globale.

Les résultats soulignent la valeur pratique du machine learning pour améliorer la fiabilité des livraisons et l'efficacité opérationnelle dans la logistique du e-commerce. Une prédiction plus précise des retards peut aider les prestataires logistiques à mieux allouer leurs ressources, à améliorer la communication avec les clients, à réduire les coûts opérationnels et à renforcer la qualité du service. Cette étude contribue également au domaine de l'analytique prédictive en supply chain management en proposant un cadre évolutif et interprétable pour la prévision des retards de livraison à partir de données logistiques réelles.

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Abbreviation List

- AI – Artificial Intelligence
- ANN – Artificial Neural Network
- API – Application Programming Interface
- CSV – Comma-Separated Values
- DL – Deep Learning
- EDA – Exploratory Data Analysis
- GBM – Gradient Boosting Machine
- KNN – K-Nearest Neighbors

- MAE – Mean Absolute Error
- MAPE – Mean Absolute Percentage Error
- ML – Machine Learning
- NLP – Natural Language Processing
- OLS – Ordinary Least Squares
- PII – Personally Identifiable Information
- RF – Random Forest
- RMSE – Root Mean Square Error
- RQ – Research Question
- SCM – Supply Chain Management
- SKU – Stock Keeping Unit
- SQL – Structured Query Language
- SVM – Support Vector Machine
- XGBoost – eXtreme Gradient Boosting

GENERAL INTRODUCTION

The rise of electronic commerce has dramatically affected logistics in the world and changed the customers' demands for deliveries. In recent times, customers shopping online have shown increased need for faster and safer deliveries. Consequently, the logistics and electronic commerce industries have been required to improve the delivery processes, especially at the last mile of delivery.

The last mile of delivery is the phase in logistics where goods are delivered from the warehouses to the consumers. Despite being geographically shorter, the last mile of logistics is considered to be the most difficult and costly part of the logistics process. Congestion, urban unpredictability, differing densities, the customers' availability and adverse weather conditions are some of the

The problem of delivery delays is one of the major problems facing e-commerce companies because delivery delays have an impact on both consumer satisfaction and the bottom line. Modern consumers not only want timely delivery but accurate delivery estimates. Failing to satisfy their demands will most likely lead to a rise in customer complaints, refund requests, and even bad consumer reviews.

Traditionally, logistics companies used static routing, averages of past data, and other conventional techniques to plan their logistics operations. Though useful for basic logistical planning, traditional techniques are unable to account for the complex dynamics of real-world logistics networks. Hence, the increasing use of predictive analysis and machine learning models to solve problems related to forecasting.

There are many benefits associated with utilizing machine learning algorithms for delivery predictions as they are able to analyze big data sets from the past and uncover sophisticated correlations between the factors. Making use of historical data from deliveries, the algorithms are capable of identifying potential delivery delays before they happen. Such features provide an opportunity for optimizing the delivery process, improving customer relations, and making the process as effective as possible.

The topic of the present paper is to explore how machine learning algorithms can be used to predict last-mile delivery delays within e-commerce operations. The research will focus on analyzing the Brazilian E-Commerce Public Dataset offered by Olist via Kaggle. Four algorithms, namely, Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor will be compared.

This research follows the approach of quantitative research methodology based on predictive analysis using a positivist research philosophy. The dataset used in this research includes about 115,000 live orders from e-commerce operations carried out during the period from 2016 to 2018

stored in different relational tables including orders, items, customers, sellers, and payments. After proper processing, a number of operational variables have been generated that could enhance the predictive accuracy.

The main aim of this research is to find out the best predictive machine learning algorithm along with an assessment of how predictive analysis could be useful in improving the efficiency in e-commerce logistics operations.

Structure of the Thesis

Three main chapters will be included in this research. The first chapter contains a complete literature review of operations involved in last-mile delivery, logistics-related difficulties, applications of predictive analytics and machine learning in supply chains, and others. The second chapter introduces the methodology which includes research design, data description, data pre-processing approach, feature engineering process, model building approach, and ethical concerns. In Chapter Three, empirical results, the analysis of performance of four machine learning algorithms through MAE, RMSE, and R^2 , the discussion of conclusions, and their implications for the logistics field of e-commerce industry will be presented.

This research will yield both theoretical and practical implications. Theoretically, this research will contribute to the understanding of the application of machine learning techniques in solving logistics prediction problem. Practically, the findings of the paper will benefit logistics and e-commerce organizations.

With the ever-growing nature of global e-commerce, there is a great need for the use of predictive logistics to remain competitive. It is for this reason that this research underscores the increasing need for incorporating data science and machine learning into logistics management systems.

Background of the Study

The fast-growing pace of the development of e-commerce and online retail platforms has resulted in dramatic changes to the sphere of logistics and supply chains during the last decade. The rapid development of digital business driven by advances in technology, shifting consumer preferences, and rising availability of the Internet have put great strain on the logistics sector to ensure that deliveries are made rapidly, effectively, and successfully. Global sales from e-commerce websites grow annually and result in rising delivery rates for logistics organizations.

In a variety of logistic procedures carried out in the course of the entire supply chain, last-mile delivery has become one of the most difficult and costly elements. This procedure encompasses

the delivery of goods from the point of distribution centres or regional warehouses to end users. Even in spite of the relatively short travelling distances, last-mile deliveries prove very sensitive to inefficiency due to various reasons including traffic congestions, non-availability of clients, weather disruptions, inefficient route planning, seasonal fluctuations, etc.

Present days' e-commerce environment calls for efficient and flexible delivery processes that can deliver thousands of goods per day to customers located at dispersed locations across the region. Moreover, today's clients are expecting increasingly faster and more transparent delivery services. The ability to provide customers with prompt delivery estimates, same day/next day deliveries, and tracking is now crucial.

The conventional delivery forecast techniques are typically based on static scheduling systems, average values, and rule-based logistics planning systems. While such techniques offer basic guidelines for logistics operations, they do not always consider the dynamic nonlinear interdependencies affecting the delivery process in real-world scenarios. The delivery delay is impacted by a variety of factors, including features of the order, freight costs, temporal purchasing behavior, dimensions of the products, geographical location of sellers, and demand variations.

The developments in artificial intelligence technologies present an innovative opportunity for delivering more precise forecasting of delivery delays in logistics systems. Artificial intelligence techniques can effectively analyze data from the operations of logistics processes and automatically detect interrelationships among relevant variables.

The rise in access to logistics data, cloud computing infrastructure, operational tracking capabilities, and machine learning libraries has driven the incorporation of predictive analytics into supply chain management. Modern e-commerce applications have enabled an extensive amount of transactional and operational data to be captured by order management systems, payment portals, delivery systems, and user engagements. Such information enables effective intelligent forecasting capabilities to enhance the efficiency and effectiveness of logistics operations.

Even though previous works have considered predictive analytics in logistics management, only a few have compared different machine learning regression algorithms for prediction of last mile delivery delays from real world big data e-commerce datasets. This work aims at bridging the above research gap by examining Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor on the Olist Brazilian E-commerce dataset.

By integrating operational, financial, temporal, and product-related variables, this research aims to develop predictive models capable of improving delivery forecasting performance in e-commerce logistics environments.

Rationale of the Study

The rationale for this study arises from several important operational and research challenges within modern e-commerce logistics systems.

Forecasting Accuracy Limitations

Despite the numerous innovations in logistics technology, delivery delay is still an issue in many e-commerce fulfilment operations. Forecasting methodologies commonly used tend to make certain static assumptions, use averages from historical data, and employ straightforward calculations. As a result, inaccuracy in forecasting can lead to inefficiency in logistics management, as well as customer dissatisfaction due to delays in delivery.

It is worth mentioning that machine learning algorithms show great promise for enhancing the quality of forecasting due to their capacity to examine large sets of historical data and find nonlinear correlations between a number of operational variables at once.

Integration of Multiple Operational Variables

Modern logistics systems consist of numerous interrelated variables that all affect delivery times collectively. The set of these variables includes seasonal buying trends, product dimensions, shipping costs, the amount of the orders, locations of sellers, and changing demands from customers. It becomes difficult to incorporate all those disparate factors into traditional analysis algorithms.

The application of machine learning techniques proves itself to be effective for processing large multidimensional logistics data since such an approach is capable not only of dealing with huge amounts of information but also revealing hidden connections and interdependencies between features.

Adaptive Learning Capabilities

The logistics system always evolves depending on the behavior of consumers, demand for products in the market, changes in urban infrastructure, and the operational strategies. Static models of prediction generally become inefficient with time since their effectiveness is dependent on constant operational conditions.

Machine learning algorithms, however, offer an opportunity for continuous improvement of the prediction models through adaptation. The capacity to adapt can be useful in an environment such as e-commerce logistics operations where things always keep on changing.

Economic and Operational Impact

Better prediction of deliveries would provide many benefits to both the logistics providers and e-commerce businesses. Predicting the actual delivery can help lower costs that arise from issues such as wrong delivery, unnecessary routing, customer service calls, and late deliveries.

The advanced forecasting tools also contribute to better consumer satisfaction because they help make the process of delivering goods more reliable and transparent. Better forecasting helps to streamline the processes and allocate resources efficiently, thus ensuring maximum efficiency.

Predictive delivery tools can also be used to make strategic decisions in the future, for example, predicting and avoiding high risk delivery practices.

Motivation of the Study

The rationale behind conducting this research is highly related to the increasingly important role of e-commerce logistics and the need for more sophisticated decision-making support systems based on data analytics. As the practice of shopping on the Internet gains greater significance and becomes one of the key aspects of contemporary consumer behavior, logistics companies must constantly seek ways to enhance the efficiency of their operations.

Indeed, last mile delivery is one of the major challenges faced by e-commerce logistics systems. Rapidly growing services such as same-day and next-day deliveries greatly increase the complexity of logistics planning and impose additional consequences on delivery failures. In other words, businesses that are unable to offer customers fast and reliable delivery services will likely lose their trust and reduce their competitive advantage.

From the technical viewpoint, recent breakthroughs in AI, machine learning, big data analytics, and cloud computing may offer innovative solutions that can help develop better forecasting systems and analyze operational data.

The financial repercussions of delays are yet another reason why the significance of the study becomes even greater. Delivery delays result in increased costs on transportation, customer service, compensation claims, and reduced efficiency. Given that the market of e-commerce is quite competitive, there are many reasons to believe that timely delivery may be a strategic tool helping organizations stay profitable and competitive.

The relevance of the study can also be attributed to an ever-growing trend of incorporating analytics and machine learning into the management of supply chains in business settings. While there is no doubt about the popularity of machine learning, especially within industry sectors, much work still needs to be done regarding the practical application of machine learning within the logistics forecast.

Statement of the Problem

However, despite considerable progress achieved in terms of logistics technologies and management, delivery delays remain one of the key problems faced by e-commerce organizations all over the world. Classical forecasting techniques employed by most of the modern logistics operations often cannot detect the intricate nonlinear relationships between the operational factors and the outcomes of deliveries.

Classic forecasting techniques include such practices as using averages from previous periods, static scheduling methods, or simple statistical techniques. Although such approaches can provide certain information that can be helpful in managing operational activities, they have a rather poor capacity to conduct analysis involving multidimensional logistics data that incorporates interaction between various factors affecting deliveries.

E-commerce logistics operations create an immense amount of structured operational data generated through transactional, tracking, and payment platforms, customer communication channels, and other systems. Unfortunately, classic forecasting techniques often do not enable effective utilization of this valuable information.

The inability to accurately predict delivery delays creates several operational and strategic challenges. Logistics providers face increased transportation costs, inefficient route planning, delayed customer communication, failed deliveries, and reduced operational efficiency. Customers experience dissatisfaction due to inaccurate delivery estimates and delayed shipments, negatively affecting customer loyalty and company reputation.

Thus, the main issue which is being discussed in this study is the difference between the complexity of modern logistic systems in e-commerce and the lack of predictability of the

traditional models used for prediction of delivery delays. Machine learning-based solutions are offered in order to solve the issue.

Objectives of the Study

The main goal of this research is to construct and analyze machine learning models aimed at estimating delivery delays in e-commerce logistics by utilizing operational databases.

The research aims at achieving the following goals:

- To explore the connection between operational variables and delivery delay characteristics in e-commerce logistics.
- To uncover the key factors which affect delivery predictions positively.
- To develop and compare several machine learning models which can be used to estimate delivery delays in e-commerce logistics, such as Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor.
- To compare machine learning algorithms with traditional statistical techniques.
- To explore how temporal, financial, and product-related variables influence delivery prediction accuracy.
- To analyze the benefits of using machine learning in the context of improving delivery forecasts for logistics companies and e-commerce businesses.

Research Questions

RQ1: Which operational variables have the strongest influence on last-mile delivery delays in e-commerce logistics?

RQ2: Which machine learning algorithm provides the highest predictive accuracy for delivery delay forecasting?

RQ3: How do temporal, financial, and product-related features affect delivery performance prediction?

RQ4: Do ensemble learning techniques such as Random Forest and XGBoost outperform traditional regression approaches in delivery forecasting tasks?

RQ5: How can predictive analytics improve operational efficiency and customer satisfaction in e-commerce logistics systems?

RQ6: What are the practical operational implications of implementing machine learning-based delivery forecasting systems?

These research questions are designed to address both the theoretical and practical dimensions of predictive logistics analytics while supporting the development of accurate and scalable forecasting models.

Scope of the Study

This paper concentrates on the creation and assessment of ML-based prediction models of the delay of deliveries in the last mile stage within e-commerce logistics operations.

Empirical analyses are performed based on the Brazilian E-Commerce Public Dataset supplied by Olist via Kaggle platform. It includes around 115,000 practical instances of e-commerce transactions gathered from 2016 till 2018 in several related datasets with the following types of information: customers' profiles, sellers' profiles, order information, products, and payments.

The investigation is mainly aimed at analyzing structured operational factors influencing the last-mile delivery process, such as:

- Purchase timestamps
- Freight values
- Product prices
- Payment values
- Product weight
- Product dimensions
- Temporal purchasing patterns
- Derived financial and volumetric features

The study focuses specifically on supervised machine learning regression techniques, including:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor

Performance evaluation is conducted using standard regression metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score.

The scope of the study is limited to historical operational data available within the Olist dataset. Real-time traffic information, weather conditions, warehouse operational metrics, and live GPS tracking data are not included in this research due to data availability constraints.

The research primarily focuses on predictive modelling and comparative algorithm evaluation rather than real-time deployment architecture or logistics optimization systems. Nevertheless, the findings are expected to provide practical insights for improving delivery forecasting accuracy and operational decision-making within e-commerce logistics environments.

Organization of the Thesis

Chapter 1: Literature Review

The present chapter provides an overview of the latest literature sources discussing the issues of e-commerce logistics, last-mile delivery, predictive analytics, and machine learning in logistics forecasting. This chapter serves to establish the theoretical foundation of the research by reviewing the history and development of logistic forecasting techniques, which evolved from the classical statistical approach towards machine learning-based logistic prediction models.

The literature review discusses such important aspects of logistics and delivery prediction as delivery delay, operational performance, customer satisfaction, quality of the logistics service provided, and supply chain optimization. Special focus was made on the last-mile problem, considering its major aspects such as traffic conditions, route planning and selection, customer presence, delivery density, and operational uncertainty.

Moreover, the literature review pays attention to the application of machine learning algorithms in logistics prediction analysis. The focus was made on such popular algorithms as Linear Regression, Decision Tree Regressor, Random Forest Regressor, XGBoost Regressor, among others, as well as their advantages and disadvantages in regard to the capacity to analyze nonlinear dependencies in the data set. A careful review of prior research allows one to outline several key areas of the research gap, which include the comparative evaluation of machine learning

Chapter 2: Methodology

The second chapter will discuss the methodology employed for carrying out the empirical analysis. This chapter will describe the research methodology, including research design, the source of data, pre-processing approach, feature engineering process, and modeling technique. The research methodology chosen follows the principles of positivism and quantitative research with a predictive analytics perspective in line with comparative objectives of the research.

The empirical analysis is done using Brazilian E-commerce Public Dataset published by Olist on Kaggle platform. In this chapter, it is explained that six related tables were brought together to create one comprehensive dataset of around 115,000 delivered orders, the treatment of missing data, transformation of dates, and feature engineering from operational variables.

Four supervised learning regression models have been designed: linear regression, decision tree regressor, random forest regressor, and xgboost regressor. The model fitting and assessment process involves splitting the data set using the 80/20 rule. Model evaluation is based on three regression metrics including mean absolute error (MAE), root mean squared error (RMSE), and R-squared.

Moreover, the tools involved in the data analysis (Python with pandas, scikit-learn, XGBoost, seaborn, and matplotlib on a Google Colab platform), ethical issues related to working with secondary anonymized data sets, and four specific limitations in methodology will be explained.

Chapter 3: Results and Discussion

This chapter discusses the empirical results that were produced by training and testing the four machine learning models with the help of the held-out test set that included 23,006 observations. This chapter focuses on comparing the performance of Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor in terms of MAE, RMSE, and R^2 , as well as explaining the reasons for the choice of the best-performing model.

This chapter will discuss why the Decision Tree yielded negative R^2 (which indicates severe overfitting); why XGBoost, which usually performs great in many settings, turned out to be the runner-up; and the implications of the empirical results for delivery prediction as a non-linear problem characterized by multiple factors.

The chapter will also focus on discussing the practical implications of the findings for logistics companies, e-commerce websites, and supply chain management. Specific recommendations about the application of predictive analysis will be offered in regard to at-risk order flagging, demand-peak identification, customer communication, and other decisions. Limitations associated with the results are identified before transitioning into the General Conclusion.

General Conclusion

The general conclusion contains information about the findings of the research and their theoretical, methodological, and managerial importance. This part describes the ways to enhance prediction of delivery delay with the help of machine learning approaches that will result in effective logistic processes in online trading.

This part provides the limitations of the conducted study. These are issues concerning the limited number of observations, excluding some operational factors from analysis, and lack of

information on weather conditions. Lastly, this chapter discusses possible directions for further research in the field of online logistic processes.

CHAPTER 1: LITERATURE REVIEW

1. Introduction

1.1 Background and Context

In recent years, the development of e-commerce has had a profound impact on the logistics industry like no other field of business. Instead of a steady supply chain system within the retail sector, today's logistics have become a dynamic business process where even the act of delivering the goods to the consumer plays a major role. It is obvious from the statistical figures showing the annual revenue earned from e-commerce transactions worldwide, which was estimated at around USD 5.2 trillion in 2021 and predicted to grow further to almost USD 8.1 trillion by 2026 (Statista, 2022).

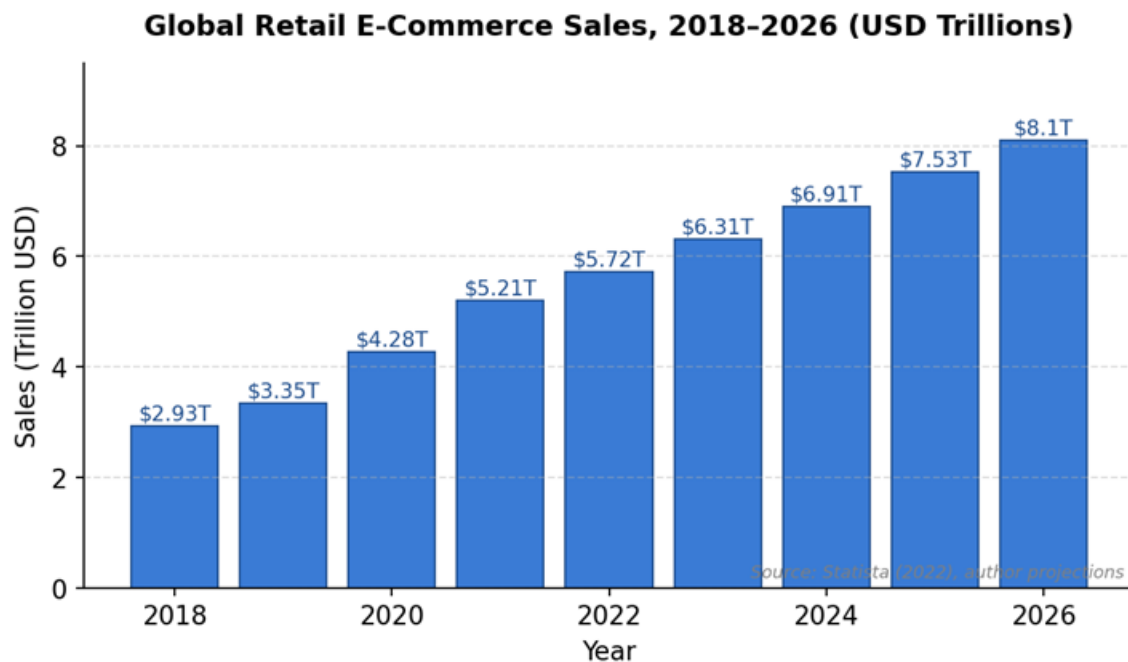


Figure 1. Global retail e-commerce sales trajectory, illustrating the demand pressure driving last-mile innovation.

In such an environment, efficient delivery has stealthily emerged as one of the strongest competitive advantages that online retailers possess. Consumers are not just comparing retailers in terms of price and the range of products offered but also in terms of how fast, reliable, and transparent the delivery process is. According to a study by Capgemini Research Institute (2019), about 88% of consumers are willing to pay more money to avail themselves of faster deliveries, such

as those that are available through next-day or same-day delivery services. In one statistic lies the problem that each retail company

However, the demands put on operational processes have grown so high that many companies are not prepared to cover their costs, yet addressing such challenges puts pressure on systems that were not designed for this level of efficiency.

Out of all the stages in the logistics chain, perhaps the last mile presents the greatest challenge for businesses, both in terms of complexity and cost. By definition, this stage means transporting goods from a distribution centre to the client's place of residence. Even though the process seems relatively simple when considered from the perspective of logistics, its execution turns out to be rather complicated due to a variety of possible obstacles. For example, traffic, weather conditions, lack of parking space, difficult street access, and unavailability of a customer may impede the process of delivery and make it a costly affair for businesses.

The Brazilian e-commerce sector, which constitutes the theoretical foundation of the current research paper, perfectly exemplifies this pattern. At the moment, the national market of online purchases is the largest e-commerce market in Latin America and one of the fastest-growing developing markets around the globe. In 2021, the online retail sales of the country reached approximately R\$161 billion, reflecting a 27% increase from the previous year (EBIT|Nielsen, 2022). As the market matures, competition between logistics providers has intensified, and delivery efficiency has become a key differentiator for retailers operating in the region.

1.2 Problem Statement

current logistic environments, as modern delivery performance is affected by many different variables. Factors such as traffic, weather conditions, order quantity and value, the destination location, delivery time, day of the week, and load at neighboring delivery centers play an important role; the way these factors interact is very difficult to predict for static delivery-planning models.

Machine learning can be a solution to this problem. It allows for learning non-obvious patterns and making forecasts based on large datasets of historical delivery performance. Predictive models can be used to identify problematic deliveries ahead of time, optimize routes, and improve delivery estimates given to customers. Machine learning has proven to be efficient in many industries, including finance, marketing, healthcare, and others; yet it remains rather unexplored when it comes to practical application in logistics, particularly when compared to different machine learning techniques. Although numerous advancements have been made in relation to technological development in logistics, there is still the issue of delivery delays. According to the McKinsey & Company report of 2020, up to 20

percent of deliveries through e-commerce portals experience delayed delivery. This number may appear insignificant at the onset.

However, its repercussions are much more serious than what they might seem to be. For instance, delayed deliveries result in increased costs since there are customer service complaints to handle, re-delivery efforts by the delivery drivers, and compensations for dissatisfied clients. Additionally, it causes a decrease in customer satisfaction since delivery delays rarely confine themselves to just one transaction. It ends up being posted on social media sites, becoming a negative review on a client portal, or even worse, the end of doing business with a certain company.

Many conventional delivery planning techniques still use static schedules or historical moving averages. While they may offer adequate rough estimates, their approach is not tailored to the complex reality of

1.3 Research Objectives and Scope

The purpose of the present review is to combine the literature regarding the issue of delay prediction in e-commerce logistics, with an emphasis on machine learning techniques. This study will explore the theoretical background behind the topic, empirical tests of theories, and modeling approaches typically adopted by researchers.

In practical terms, the review concentrates on three closely connected areas:

- The operational characteristics and challenges that make last-mile delivery uniquely difficult.
- The factors—operational, geographic, temporal, and human—that drive delivery performance in e-commerce settings.
- The role of machine-learning techniques in building predictive models for logistics and supply-chain management.

By working through these three threads, the review builds a conceptual foundation for the current study and identifies the specific gaps that this research aims to address.

1.4 Structure of the Literature Review

The rest of this literature review will be structured into eight parts. Following this introduction, Part 2 provides an overview of the theoretical underpinnings and frameworks that provide context for the study of logistics performance and the application of predictive analytics. Part 3 highlights the operational aspects that contribute to the complexity of last-mile deliveries in the context of e-commerce. Part 4 discusses the variables affecting delivery performance, including those related

to orders, as well as other external factors, such as seasonality. Part 5 deals with machine learning by tracing its development in logistics analytics and identifying the algorithms most frequently utilized in the current state-of-the-art practices. Finally, Part 6 offers a summary of relevant findings from studies comparing different types of predictive models.

2. Theoretical Foundations and Conceptual Frameworks

2.1 Supply Chain Management Theory

From supply chain management theory we can gain insights into the issues related to logistics processes and delivery performance. In their work, Lambert and Cooper (2000) define supply chain management as the coordination and integration of critical business processes from raw material suppliers, through manufacturing and warehousing, to the end customers. One should note that according to this definition and generally speaking, the key factors in keeping a chain going are coordination and information sharing.

Thus, looking at the problem under consideration through the lens of supply chain management theory, we can conclude that delivery performance cannot be regarded as an independent process or even an event. Rather, it becomes the obvious result of multiple prior steps made, including scheduling of procurements, positioning of inventories, designing of warehouses, planning of routes, and customer service arrangements for handling exceptions.

The Resource-Based View of the firm (Barney, 1991) provides an additional approach to the discussion. The concept suggests that organizations secure competitive advantage through having resources that are valuable, rare, and difficult to imitate. With contemporary logistics, data-analytic capabilities certainly fit into this category. Organisations capable of acting on the results obtained from the analysis of operational data enjoy advantages over those which are not able to use their potential.

Hunt and Davis (2012) develop this idea further by using Resource-Advantage theory. It assumes that knowledge and information can become engines of competitive advantage for organisations. This means that, in logistics, advanced data analysis and machine learning is much more than just technical aspects that provide benefits to organisations.

Dynamic Capabilities Theory (Teece, Pisano, & Shuen, 1997) is another crucial theory in the context mentioned above. According to this particular theory, the need for resource reconfiguration arises from the requirement to react to changes that occur in the environment. As far as logistics is concerned, the notion of reconfiguration capability becomes apparent in connection with fleet

management since there should be capabilities to change the routes of the fleet, increase the number of drivers for the peak time or replace the technology used if it fails.

Table 1. Theoretical frameworks anchoring delivery-performance research.

Framework	Key Authors	Core Idea	Relevance to Delivery Prediction
Supply Chain Integration	Lambert & Cooper (2000)	Integrated cross-organisational processes	Delivery as outcome of upstream coordination
Resource-Based View	Barney (1991)	Rare and hard-to-imitate resources drive advantage	Data and analytics as strategic resources
Resource-Advantage Theory	Hunt & Davis (2012)	Knowledge and information enable superior performance	ML capability as a competitive asset
Dynamic Capabilities	Teece et al. (1997)	Firms adapt resources to changing environments	Adaptive routing and learning-based systems
Complexity Theory	Choi et al. (2001)	Non-linear interactions in complex adaptive systems	Logistics networks behave as CAS
Systems Thinking	Sterman (2000)	Performance arises from system structure	Holistic modelling of delivery outcomes

2.2 Service Quality Theory in Logistics

Service quality is another important criterion that can be considered, mainly because customers will remember an unreliable service much longer than reliable one. According to Parasuraman et al. (1988), there are five major dimensions of service quality, which include reliability, responsiveness, assurance, empathy, and tangibles.

Later, the concept was further extended by Mentzer et al. (2001) who developed the Logistics Service Quality (LSQ) construct. In this context, nine major factors were found to affect the way the service quality is perceived. They include order accuracy, information quality, timeliness, order condition, discrepancy handling, etc

Among all mentioned factors, the most important one seems to be reliability of deliveries. In other words, it is a question of how timely and in what condition shipments reach their destinations. As shown by Bienstock et al. (1997), reliability significantly affects customer satisfaction and repeat business rates.

The Gap Model of Service Quality (Parasuraman et al., 1985) takes this one step farther by focusing on where the problem lies with regard to service quality – in the gap between customer expectations and reality. When it comes to logistics, the usual occurrence is the gap between the

delivery date promised and the actual receipt of the package. Predictive analytics may take care of that without anyone knowing, by delivering a more realistic and detailed estimation of when something will arrive.

Table 2. Logistics Service Quality dimensions (Mentzer, Flint, & Hult, 2001).

Dimension	Description	Example in Delivery Operations
Order accuracy	Match between order placed and order received	Correct SKUs, quantities, and configurations
Information quality	Availability and richness of order information	Accurate tracking and ETA notifications
Timeliness	Delivery within promised time window	Arrival within a 2-hour slot
Order condition	Physical state on arrival	Undamaged packaging and product
Discrepancy handling	How issues are resolved	Speed of refund, redelivery, or replacement
Personnel contact	Quality of interactions with staff	Politeness and competence of drivers
Order release quantities	Availability versus what was ordered	Avoiding partial shipments
Order quality	Conformance to expected specifications	Right product, right configuration
Ordering procedures	Ease of placing the order	Smooth checkout and clear delivery options

2.3 Information Processing Theory

Another useful approach for understanding logistics is Information Processing Theory. According to Galbraith (1973), organisations must be able to process information in accordance with the environment they operate within. The contemporary logistics landscape is rich in information.

A logistics facility produces huge amounts of information continuously, including such details as orders, shipments, GPS signals, deviances, traffic conditions, weather forecasts, and customer feedback. Manual processing of all this information is virtually impossible, which is the reason why automated analysis tools have become a necessity rather than an asset.

Moreover, there is the problem of information asymmetry that is not widely recognised but that is important to the customer side of the equation. Typically, consumers receive very limited information about the process of delivering goods: an order confirmation and a couple of tracking alerts are all that most people get before receiving a package. Here is where predictive analytics helps alleviate the problem of asymmetry by giving consumers realistic expectations through the power of estimation.

2.4 Complexity Theory and Systems Thinking

In this regard, Information Processing Theory can also help gain a better comprehension of logistics activities. According to Galbraith (1973), the organisation should be capable of processing the relevant information based on its surrounding environment. The modern logistics environment is abundant in information

The logistics facility generates tons of information on an ongoing basis, which may include such data as orders, deliveries, GPS signals, deviances, traffic conditions, weather forecasts, and consumer feedback. It is quite impossible for a person to manually process all the available information, hence the necessity to use advanced analysis tools in this context.

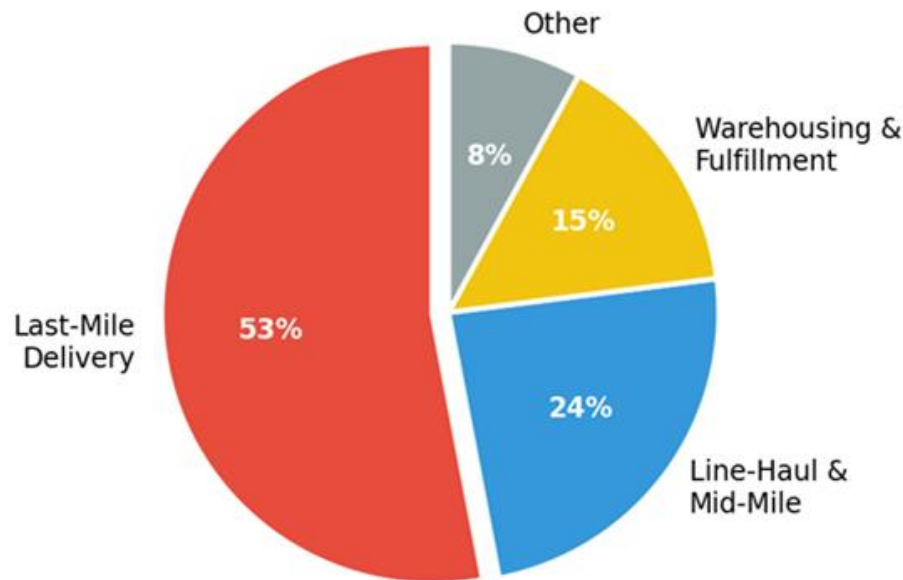
It is also worth considering the issue of informational asymmetry that is often neglected but important for the customer's perspective. In general, consumers receive minimum information related to their orders: usually only order confirmation and some tracking notifications prior to actual delivery of the item. This challenge may be solved using predictive analytics, which allows consumers to set realistic expectations via estimation.

3. Last-Mile Delivery Challenges in E-Commerce

3.1 Operational Characteristics of Last-Mile Delivery

Last mile delivery is the final stage, and possibly the toughest one, of the supply chain process. This refers to the stage whereby products move from their last point of storage to the consumer's location. According to Boyer et al. (2009), the last mile can best be described as being the costliest, least efficient, and environmentally unfriendly section of the supply chain process. This statement is uncomfortable but truthful considering the fact that the last mile constitutes up to 53% of total shipping costs (Business Insider, 2021).

Share of Total Shipping Costs by Stage



Source: Business Insider (2021); Boyer et al. (2009)

Figure 2. Share of total shipping costs across supply-chain stages, highlighting the dominance of the last mile.

Several features make last-mile delivery uniquely difficult. The first is density. Residential areas typically have much lower delivery density than commercial zones, which means drivers cover more ground for each parcel dropped. Customer availability is another wild card. Even with perfect routing, an empty house turns a planned delivery into a failed attempt and forces costly rework. Urban congestion, narrow streets, and disappearing parking spaces further slow operations down. And the sheer variety of delivery locations—apartments, gated communities, offices, parcel lockers, pickup points—forces operators to support multiple modes of last-mile service in parallel.

Gevaers, Van de Voorde, and Vanelander (2014) propose a useful taxonomy of last-mile challenges, grouping them into five broad categories: consumer-related factors (availability and accessibility), operational factors (routing and scheduling), regulatory factors (traffic restrictions and emissions standards), infrastructural factors (parking and road conditions), and technological factors (tracking systems and communication tools). What is striking about this taxonomy is the degree to which these factors interact, which is precisely why prediction and optimisation in the last mile are so demanding.

Table 3. Five categories of last-mile delivery challenges (Gevaers, Van de Voorde, & Vanelander, 2014).

Category	Examples	Typical Impact	Mitigation Strategy
Consumer-related	Customer availability, address quality	Failed deliveries, re-attempts	Time-window booking, parcel lockers
Operational	Routing, scheduling, fleet management	Higher cost per drop, longer routes	Dynamic routing, ML-based prediction
Regulatory	Low-emission zones, time-of-day rules	Limited delivery windows, vehicle changes	Electric fleets, off-peak deliveries
Infrastructural	Roads, parking, building access	Slower drops, congestion	Micro-hubs, walking couriers
Technological	Tracking, communication systems	Information asymmetry, customer queries	Real-time tracking, predictive ETAs

3.2 Urban Logistics and City Distribution

Cities make everything about the last mile difficult. According to Taniguchi and Thompson (2014), "city logistics can be defined as an activity of optimization of transportation and logistics related activities in urban environment in consideration of factors such as congestion, safety, energy, and environmental concerns." Although urban settings provide opportunities for efficiency because of the density of the customer base, they also accentuate the costs associated with delays caused by congestion or missed opportunities to deliver products due to inefficiency in a curb side stop.

As Figliozi (2010) discovered, congestion might extend delivery times by up to 40% during peak times. Moreover, congestion is neither consistent nor stationary – it changes on an hourly, daily, or even per-street basis, which makes it difficult to account for using the usual rule-based approach to scheduling. In this regard, machine learning models, based on real-time traffic data, might outperform more traditional algorithms

Another challenge that complicates the issue even further stems from environmental regulations. Many cities today impose low emission zones and delivery time restrictions and prohibit vehicles above certain sizes in order to reduce pollution (Browne, Allen, & Leonardi, 2011). These policies are mostly well intentioned, but they raise the operational bar by requiring delivery planners to satisfy multiple constraints at once. Advanced analytics tools that can balance cost, time, and emissions simultaneously are becoming essential rather than optional in such environments.

3.3 Customer Expectations and Service Requirements

E-commerce has changed consumer expectations regarding delivery, usually at a faster pace than logistics systems have been able to adapt. According to Nguyen, de Leeuw, and Dullaert (2018), expectations related to speed, reliability, flexibility, and visibility are the primary ones. Additionally, the 'Amazon effect' has helped in establishing expectations for quick and free shipping. As per Convey (2020), 63% of people expect that their standard shipments will reach them within three days.

Narrow time windows constitute yet another issue. Agatz et al. (2011) examine the attended home delivery problem wherein the delivery cannot be completed without the presence of the customer on site. It is evident that narrow time windows improve customer satisfaction but decrease flexibility and raise the cost of operations. Thus, there is a fine line that must be maintained operationally.

Undeliveries subtly reduce logistics effectiveness. Undeliveries happen if the customer is not present or due to missing or inaccurate addresses. According to Iwan et al., undelivery ratios range from 5 to 15 percent. An undelivered package adds 20 percent to the total cost of delivery and decreases customer satisfaction levels, which explains the current interest in predicting potential undeliverable stops before the driver leaves the distribution center.

3.4 Technological Innovations in Last-Mile Delivery

There are several innovative uses of technology in last mile logistics. Autonomous vehicles, drones and robots have all been mentioned as revolutionary technologies in last mile logistics. However, as noted by Joerss et al., there are regulatory, logistical, and social challenges that prevent rapid deployment and adoption of those technologies.

A completely different path for innovation is provided through crowdsourced delivery. For example, companies such as Uber and DoorDash allow to leverage crowdsource delivery to respond to variable consumer demand. In particular, according to Archetti, Savelsbergh & Speranza, crowd-sourced delivery allows to cut costs of delivery without jeopardizing service quality.

Yet another innovation is associated with smart lockers and pickup points. Deutsch & Golany demonstrated that an optimally placed network of pickup points could decrease cost of last mile deliveries by up to 60%, yet customer acceptance plays a pivotal role in their success.

4. Factors Influencing Delivery Performance

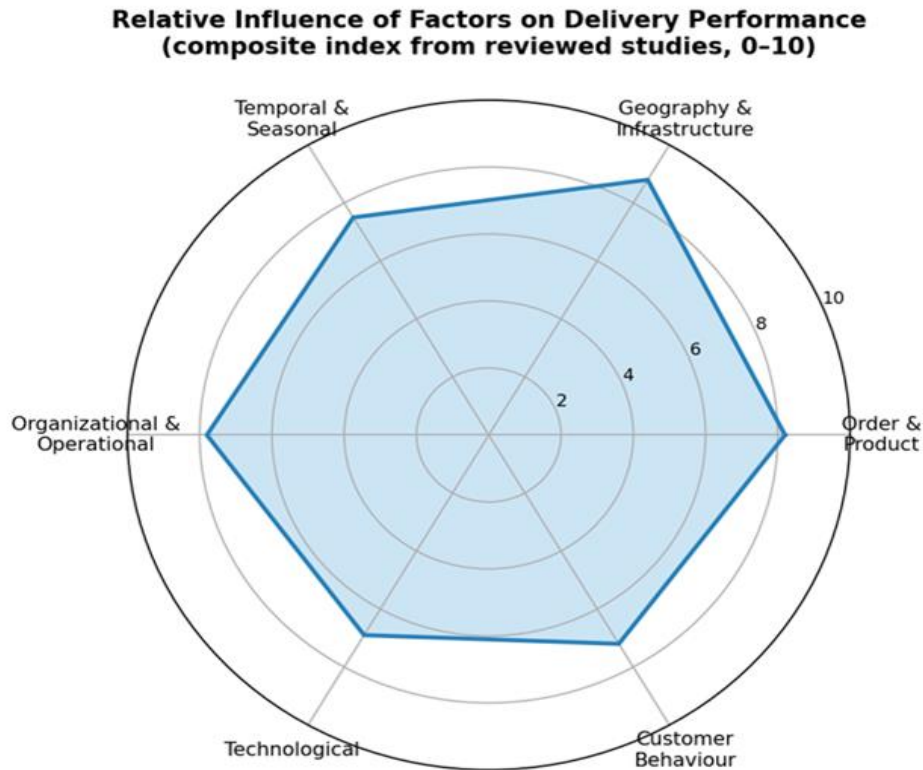


Figure 3. Composite radar of the relative influence of major factor groups on delivery performance, synthesised from the reviewed literature.

4.1 Order and Product Characteristics

The importance of order-specific details is higher than one might initially think. According to Park and Regan (2004), large orders may require special procedures, several deliveries, or even additional security in case of valuable deliveries. Such small details may influence the amount of time needed for the delivery of a single package and, when multiplied by thousands, they may have a significant impact on the overall delivery process.

Product-related factors include weight, size, fragility, and other parameters that define transportation requirements. As explained by Winkenbach and Janjevic (2018), diversification increases the level of operational complexity and raises delivery costs. Another factor that must be considered is the perishability of certain products, which implies a stricter schedule due to their depreciation during transit.

Order consolidation means merging several orders into a single shipment. On the one hand, such practice can significantly increase delivery efficiency. On the other hand, it contradicts the customer's requirement of timely deliveries. To identify the optimal approach to order consolidation, Archetti and Speranza (2017) analyze its relationship with demand dynamics and customer service levels.

4.2 Geographical and Infrastructure Factors

Geography is one of those subtle variables which help determine possibilities in the process. Distance between a DC and its customers, density of population in the region and geography between these two points affect delivery time and costs. Gonzalez-Feliu, Semet and Routhier (2014) take account of these variables when building their prediction models.

Similarly, the quality of the local infrastructure affects driving speeds and choice of routes. In case of countries like Brazil, variations in the quality of infrastructure are significant enough for national-level strategic planning where predictive modeling can play an important role.

Then again, spatial clustering cannot be ignored. According to Campbell and Savelsbergh (2005), density of deliveries is one of the key factors influencing costs in the last mile of delivery. However, demands made by online shopping do not form clusters as in the case with other retail operations.

4.3 Temporal Factors and Seasonality

Timing of the day, day of the week, and season of the year all have an impact on delivery efficiency. According to Köhler and Strehler (2015), the variation of delivery time may be highly irregular during the course of the day due to changes in traffic conditions, availability of customers, and driver's efficiency. In other words, a single route can become totally different from morning till evening.

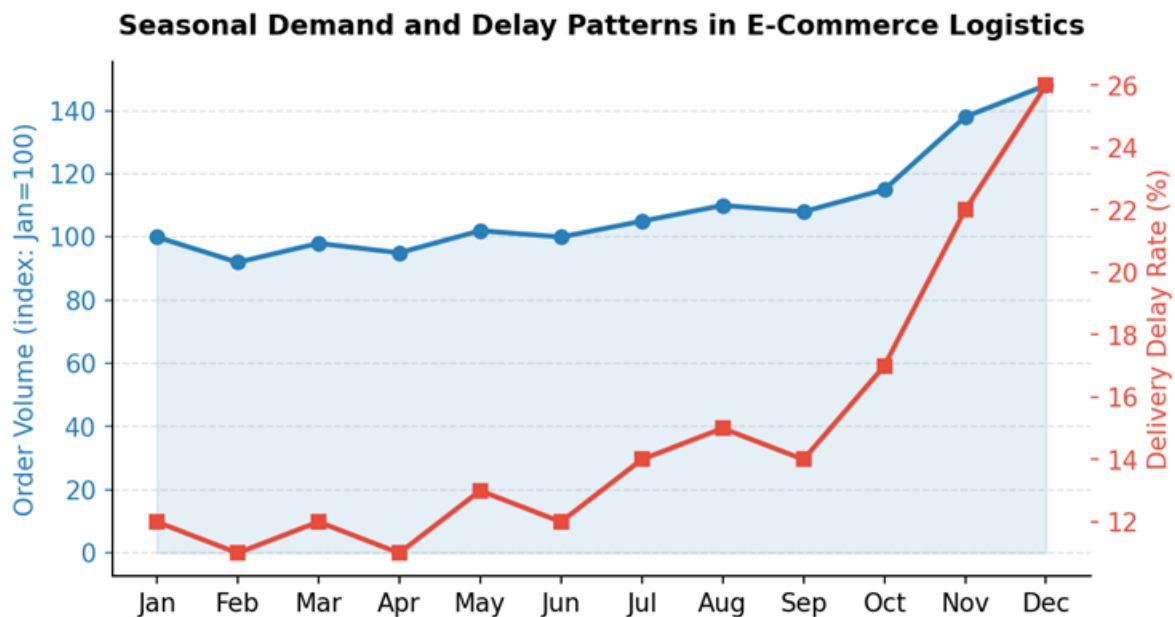


Figure 4. Seasonal demand and delay patterns showing how peak periods stress logistics capacity and inflate delay rates.

The concept of seasonality further complicates things. In times of heavy workload – like holidays or large promotions – the number of deliveries increases by 30-50% (DHL, 2020). Seasonality means additional

stress on logistical processes, which leads to an inevitable increase in the likelihood of delays. Hence, any predictive model must take into account the annual cycles of activity and remain adaptive to accommodate for unforeseen events like a new viral craze or unexpected promotion.

Weekly patterns should not be ignored either. The number of deliveries at residences will go up on weekends due to the presence of clients at home; deliveries at business addresses will be higher in the working days. The interaction between these patterns and other factors is surprisingly diverse, even within the timeframe of seven days.

4.4 Organisational and Operational Factors

Organisational capability within the logistics firms themselves influences performance. The size of the fleet, technology used, and the training given all influence this performance. Variability in the delivery performance of different service providers has been found by Yu and Egbelu (2008), and many of the reasons for this variability lie in differing organisational capabilities.

The operations of the warehouses from which orders are dispatched also impact last mile logistics performance. Efficient order processing and picking, and loading, determine how fast an order can be sent out. Boysen, de Koster, and Weidinger (2019) point out that effective coordination between warehouse and transport operations is necessary; if not, the warehouse becomes the bottleneck that slows down all other processes.

The experience of the driver cannot be underestimated either. Experienced drivers gain tacit knowledge regarding particular routes, neighborhoods, and customer needs, resulting in faster deliveries. Driver turnover rates within such companies tend to be relatively high, and therefore it can be difficult to retain knowledge. This makes driver training and support particularly valuable.

5. Machine Learning Applications in Logistics and Supply Chain Management

5.1 Evolution of Analytics in Logistics

Analytics in logistics has evolved in different phases. The first phase is descriptive analytics, which involved reporting and performance management in the past to give managers visibility of events that have already occurred. Diagnostics analytics sought answers to the reasons behind these outcomes. Most recently, machine learning-based predictive analytics enables organizations to predict events before they occur, while prescriptive analytics recommends appropriate actions.

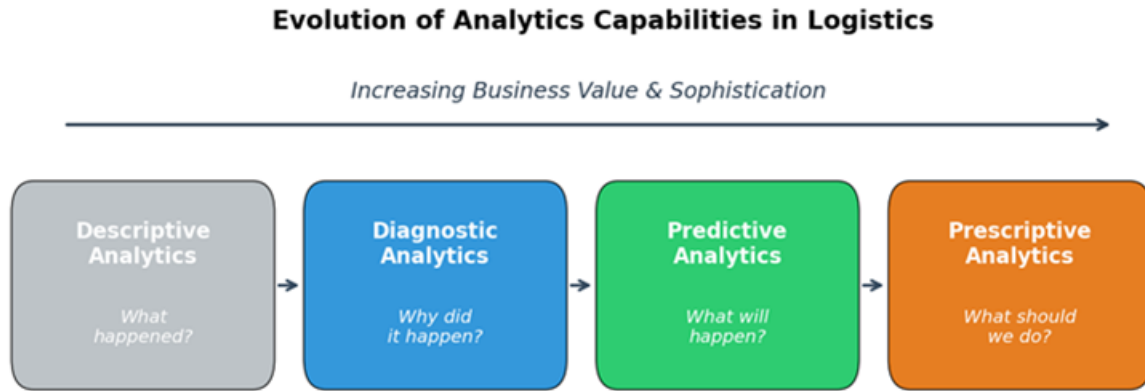


Figure 5. The four stages of analytics maturity and the operational question each stage aims to answer.

According to Wang et al. (2016), the development of logistics analytics entails the move from traditional operations research to machine learning algorithms. While optimization algorithms remain essential for cases where the problem is structured, machine learning algorithms appear more powerful when it comes to spotting patterns in complex and high-dimensional databases. In reality, machine learning and optimization are not competing methods, but, on the contrary, can be combined to achieve even greater results by integrating predictions with optimization.

The appearance of Big Data in logistics opens the door to many challenges and opportunities. Zhong et al. (2016) identify the following characteristics of logistics data as volume, velocity, variety, and veracity. Machine learning algorithms can effectively analyze such datasets due to their ability to identify non-linear associations and interaction between different variables.

5.2 Supervised Learning for Delivery Prediction

Supervised learning techniques are the main tools in delivery time forecasting. These systems are trained by analyzing the data collected previously where the output variable (like delivery time) is already known, allowing these systems to predict future outputs based on new inputs. The choice of algorithm depends on various parameters including the type of input data, accuracy requirements, computational capabilities – basically making a compromise between theory and practice.

Regression model remains popular as a basic approach. According to Ramanathan (2010), regression analysis is used by the author to analyze variables affecting delivery performance in logistics. Regression is simple and fast; however, it works poorly in cases of non-linear relationship – quite common in logistics.

Tree ensembles are gaining popularity as they can deal with nonlinearity and different types of variables easily. The Random Forest algorithm developed by Breiman (2001) is a combination of several decision

trees designed to increase accuracy and avoid overfitting. The algorithm is able to tolerate missing data, and it has relatively low sensitivity to the scaling of input variables.

XGBoost by Chen and Guestrin (2016) has demonstrated consistent high performance in a variety of logistics problems. XGBoost creates the next decision tree based on the residual errors from previous trees; hence, it learns iteratively to compensate for the mistakes. This iterative process makes it capable of detecting intricate structures in data.

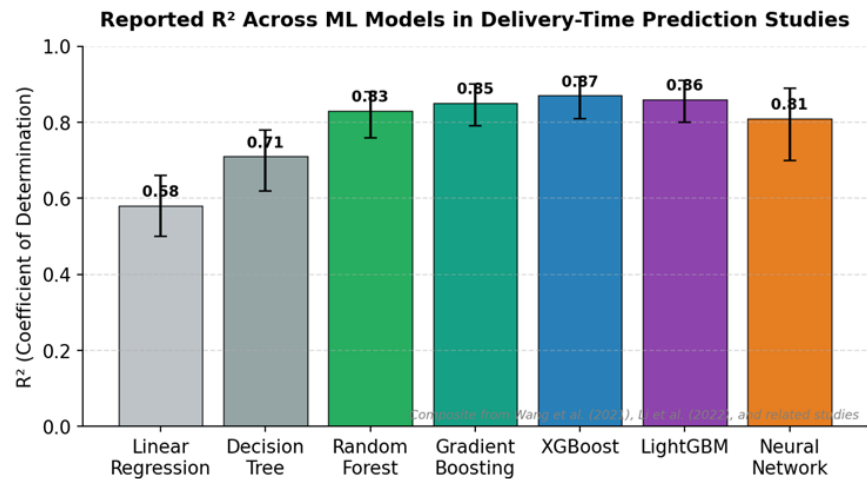


Figure 6. Reported R^2 across machine-learning models in delivery-time prediction studies. Error bars represent the typical range across reviewed studies.

Table 4. Comparative properties of common ML algorithms used in logistics prediction.

Algorithm	Strengths	Limitations	Typical Use in Logistics
Linear Regression	Simple, fast, interpretable	Misses non-linear effects	Baseline for delivery-time estimation
Decision Tree	Visual, handles mixed types	Overfits easily	Quick exploratory modelling
Random Forest	Accurate, robust to noise	Less interpretable	Delay classification & ETA prediction
Gradient Boosting	High accuracy on tabular data	Sensitive to tuning	Demand forecasting, ETA refinement
XGBoost	Strong with structured data	Computationally heavier	State-of-the-art benchmark
Neural Networks	Captures complex non-linearity	Needs large data, less transparent	Spatio-temporal demand patterns

5.3 Feature Engineering and Selection

It's often here where all the magic happens, no matter what kind of algorithm lies beneath. For applications in logistics, raw data is seldom self-explanatory, needing to undergo some transformation into features that have meaning behind them and can thus provide insights to the model. Features like time of day, day of week, season, or other temporal factors can help account for time-related patterns.

Spatial features, on the other hand, will include distance metrics or density, and geographic grouping of data. Interaction features can reveal connections not evident in any single feature alone. For example, the interaction between delivery cost and distance could indicate a certain way that a logistics provider decides to prioritise its services depending on available capacity. While domain expertise will almost always guide this task, automation techniques like polynomial or basis functions might be of assistance.

Feature selection then helps thin out the variables that are noise rather than signal. Filter methods—correlation analysis, mutual information—score features independently. Wrapper methods such as recursive feature elimination evaluate subsets of variables based on actual model performance. Embedded methods, like LASSO regularisation, perform selection as a side effect of model training itself, which is often the most efficient route in practice.

5.4 Model Evaluation and Validation

The usefulness of a beautiful engineering of a model relies greatly on the evaluation process employed. Model performance estimation using cross-validation is common practice. One of the methods used for model evaluation is the k-fold method where the dataset is divided into multiple folds for testing purposes, thus giving a better model performance estimation than the traditional train-test set.

The most realistic option to consider when evaluating the models for logistic predictions would be temporal cross-validation as it reflects real life. Using this technique, the models are evaluated using the oldest data in their training set with more recent ones being used in their test set. This is because in real life, the data from the past helps to predict the future, and not vice versa.

There are several metrics used in assessing regression models. MAE measures the absolute size of the errors while RMSE does the same but gives weight to bigger errors. R-squared is used in determining the proportion of the variance explained. There is a need to choose a metric based on business objectives and type of error.

Table 5. Common evaluation metrics for delivery-time prediction.

Metric	Formula	Interpretation	When to Prefer
MAE	$\Sigma \hat{y}-y / n$	Average absolute error in same units as target	Balanced view of error magnitude
RMSE	$\sqrt{(\Sigma(\hat{y}-y)^2 / n)}$	Penalises larger errors more heavily	When large delays are very costly
R ²	$1 - \Sigma(\hat{y}-y)^2 / \Sigma(y-\bar{y})^2$	Variance explained by the model	Comparing relative model fit
MAPE	$\Sigma \hat{y}-y / y \times 100/n$	Average percentage error	When relative error matters more than absolute

Interpretability is another critical topic in logistics models. It is not only enough to know the output of the model but also understand why the output came up in the first place. Lundberg & Lee (2017) suggested using SHAP (SHapley Additive exPlanations) values as a method to explain how individual predictions come about in a manner understandable enough for management and workers alike. Feature importance provides a broader perspective on the variables that matter across a whole dataset.

6. Comparative Studies of Machine Learning Models in Logistics

6.1 Comparative Performance Analysis

A comparative study serves as a real test of different algorithms on real-world data. Comparing performance metrics for a variety of algorithms run on the same dataset makes for a better understanding of what works in particular cases. For example, according to the research of Wang et al. (2021), ensemble methods are found to be better at estimating delivery times compared to individual machine learning algorithms. However, this is true only because the difference in results was considerable; yet, the improvement is highly contingent on specific circumstances, proving once again that there is no universally 'better' algorithm.

This statement is supported by Wolpert's (1996) no-free-lunch theorem stating that no algorithm is better than the other for solving all the problems existing. This idea has a lot to do with reality in logistics because of an extremely varied range of situations that can occur within different organizations, markets, and delivery modes.

The use of ensemble learning techniques can result in even greater accuracy. One such technique is stacking which implies using a meta-model to aggregate predictions made by several base models. The advantage of voting-based ensembling consists in its ability to balance various approaches used in individual base learners, thus providing model's weaknesses.

6.2 Computational Considerations

Efficiency is a key concern once the model is moved out of the laboratory into the real-world logistics process. Training and prediction speed and memory usage will play a part in selecting the right algorithm. Simple algorithms like linear regression and decision tree training are fast and hence appealing, despite their relative lack of predictive power, for applications that require quick turnaround. On the other hand, while ensemble models will yield greater accuracy, they will also need much more computation, which is not always possible on the edges of a live network.

Scalability is another important aspect, albeit one that is quite different from efficiency. Some machine learning algorithms can be effectively parallelised, allowing for distributed processing. Other models, however, depend on inherently sequential computations, and thus scaling up is difficult. Thus, companies have to consider both aspects – accuracy and suitability – when selecting models for use in practice.

The model-updating approach is yet another factor to consider, and this is easy to overlook. In the case of online learning algorithms, they can keep updating themselves as soon as new data comes in, thus helping them respond effectively to changes without requiring a full-fledged training process again. On the other hand, batch-learning models will require regular training as soon as they have accumulated enough data, and this will be easier and faster, though the response time to changes may take a while.

6.3 Domain-Specific Adaptations

Model adaptation based on logistics-specific features will often yield disproportionately positive results. A good example is cost-sensitive learning, where, for instance, in the case of delivery predictions, a late delivery carries a higher operational and reputational cost than an early one. Hence, symmetric loss functions can sometimes lead researchers to use a less effective model. On the contrary, asymmetric loss functions can provide better results by penalising late predictions accordingly.

Hierarchical modelling techniques may also come in handy since such an approach is capable of taking into account the hierarchical nature of logistics problems. By recognising multiple levels, multi-level models are able to capture different types of dependencies between observations as well as patterns that cannot be identified otherwise.

The last type of adaptation to consider is transfer learning. Transfer learning implies that an already-trained model can be adapted for another market or used in another situation by providing it with new training data. Thus, models do not need to be retrained from scratch, saving time and effort, especially when dealing with fast-evolving markets and limited historical data.

7. Research Gaps and Opportunities

7.1 Methodological Gaps

Despite the advancement of logistics analytics, there are still some research problems that need to be addressed. For example, most logistics analytics papers concentrate only on optimizing for one key parameter instead of addressing several simultaneously. However, the reality is that logistics specialists should have several optimization targets at once; for instance, they might aim at reducing the cost, speeding up deliveries, and ensuring reliable operations. Machine learning approaches to this problem are yet to be developed.

The next problem that deserves mentioning relates to the estimation of uncertainties. Namely, almost all prediction models offer a deterministic answer in terms of the expected delivery time without specifying whether the model is quite sure or uncertain regarding its predictions. It could be beneficial both from the risk management perspective and the customer satisfaction one if the models provided probabilistic outputs.

The third gap is one of causation. Machine learning models are highly adept at predicting things, although they usually do not go as far as determining the cause behind such predictions. In order to improve processes, causation is important because otherwise, we cannot solve the problem if we don't know the causes.

7.2 Data and Application Gaps

There are additional issues that must be accounted for in real-world data for logistics applications. One major issue would be the case of missing data, especially for failed and canceled deliveries. The lack of data itself is significant and if not managed properly may lead to bias within predictive models. The imputation techniques as well as native handling algorithms for missing data should receive further attention from the literature.

Another issue in logistics is that of class imbalance. Specifically, for many datasets concerning the delivery process, delayed deliveries form a smaller portion of the sample when compared to non-delayed deliveries. For such skewed class distributions, standard evaluation measures may give overly optimistic results due to the occurrence of rare events. However, there are solutions to such scenarios like SMOTE, cost-sensitive learning, and other evaluation metrics which require further investigation.

Perhaps the least recognized issue is concept drift. The delivery profile will change over time, either slowly because of gradual demographic change, or suddenly because of a new rule or behavioral change among consumers. Models trained on old data will become less accurate because their context will evolve over time. The use of adaptive learning models that detect and compensate for this phenomenon presents a wide-open research avenue.

7.3 Integration and Implementation Gaps

The second gap that may not always be present in scientific literature but could benefit from further study pertains to the actual logistics process involved in implementing machine learning models. Most logistics firms continue to use legacy information systems which were not developed for predictive purposes. In-depth studies into system architecture and implementation patterns may prove valuable in addressing this problem.

Human-AI collaboration represents another angle at this problem, as human beings are required to interpret the results produced by AI. It means that both explainable AI technology itself as well as human-centric design can be useful here. It also involves issues related to training and familiarization with the new technology.

Finally, ethics cannot be overlooked. The questions related to fairness and customer privacy in automated deliveries cannot remain unanswered. Therefore, developing an ethics framework related to the use of AI in logistics would represent a valuable research direction.

7.4 Comparative Analysis Gaps

Whereas there are numerous studies that have examined different algorithms, comparative studies that look at different datasets, contexts, and operational environments remain rare. Most of the research carried out has only considered one dataset, making the conclusions drawn not easily generalizable. Benchmarking datasets and establishing a framework for evaluations would go a long way towards facilitating comparisons of studies and the building of knowledge in this field.

More work needs to be done to establish the interaction between traditional statistical analysis and contemporary machine learning approaches. Understanding when an approach is too simple for the job and when increased complexity adds value is key in helping logistics enterprises make sound decisions regarding implementation.

Finally, cross-domain learning deserves attention. It is likely that solutions used in predicting traffic, estimating demand, and related operational research tasks may apply equally well to solving the last mile delivery challenge.

Table 6. Summary of identified research gaps and example future directions.

Gap Category	Specific Limitation	Promising Direction
Methodological	Single-objective optimisation	Multi-objective ML frameworks
Methodological	No uncertainty quantification	Bayesian and conformal prediction

Methodological	Limited causal reasoning	Causal ML and counterfactual analysis
Data	Missing values & class imbalance	Robust imputation, SMOTE, cost-sensitive learning
Data	Concept drift over time	Online and adaptive learning systems
Implementation	Legacy system integration	Service-oriented MLOps architectures
Human–AI	Explainability for operators	SHAP, LIME, user-centred dashboards
Ethics	Fairness and transparency	Responsible AI governance for logistics
Comparative	Single-dataset studies	Benchmark suites and protocols

8. Synthesis and Conclusions

8.1 Summary of Key Findings

The current literature review of research in the field of logistics in e-commerce regarding predicting delay of deliveries has been conducted. There are several main ideas arising from this literature analysis and guiding this research further.

Firstly, the last mile of the logistics process in e-commerce appears to be the most important and challenging stage of the whole process. In the context of delivery delay in e-commerce, its importance lies in the fact that it includes numerous challenges such as dispersed customers' locations, urban problems, and other issues, which make accurate predictions difficult using regular techniques

Secondly, delivery performance is affected by many different factors including order parameters, geographical location, time-related issues, organisational capabilities and technology. Relations between these factors tend to be rather complicated, involving non-linear effects which make the use of machine learning methods particularly relevant.

Thirdly, machine learning models have been found to provide great accuracy when predicting delivery delay in e-commerce. Tree-based algorithms and ensemble approaches, namely random forest and xgboost, in particular, are often employed for this purpose. However, the applicability of the chosen model depends greatly on the situation.

8.2 Implications for Current Research

The conclusions reached in this literature review have direct practical implications for the present study conducted based on the e-commerce data of Brazil (dataset 'Olist').

Methodology: The choice of methods in the present work is in line with the best practices identified in literature, and allows comparing three models with regard to model complexity, interpretability and predictive power.

Feature engineering: Previous research highlights the importance of using a large number of features obtained from various sources in order to be able to incorporate information about order characteristics, product, geographical location, etc. The Olist database contains precisely the data required for model building.

Evaluation: The choice of multiple criteria (MAE, RMSE and R-squared) reflects the common opinion in literature regarding the use of multiple criteria to measure model's performance. In addition, the use of multiple criteria helps build a clearer picture of the performance of each model and thus allows choosing the most appropriate one.

Practical Relevance: Real world data usage helps solve the problem of data limitations discussed in previous papers. Using data for Brazil, as an emerging market, expands the geography of empirical research in this field.

8.3 Theoretical Contributions

Theoretical contributions include three inter-related aspects. First, this research extends the use of machine learning models in logistics applications in terms of its ability to detect realistic operations patterns. Second, this paper builds on previous research on service quality by providing further insights into the variables affecting delivery reliability. Lastly, this research contributes to the literature on predictive analytics by testing several machine learning models in the context of one real-life logistics setting.

This research is also relevant to the emerging topic of data-driven supply chain management. This paper contributes to the field of data-driven supply chain management in terms of demonstrating the practical relevance of the predictive analysis approach.

8.4 Practical Implications

These findings have tangible real-world implications for various stakeholders who are a part of the e-commerce network. Logistics companies could use accurate prediction models to proactively decide on alternate delivery routes and reallocation of resources. Doing so would not only be cheaper but would enhance service quality.

E-commerce sites, with better delivery forecasts in their hands, would be able to communicate clearly with their clients, resulting in less disputes regarding deliveries

On the customer side, better accuracy means easier anticipation of when goods will arrive and fewer inconveniences in general which lead to improved customer experience and greater satisfaction in general.

Policy-makers could rely on knowledge about what factors determine good delivery performance to create better policies aimed at improving the last mile problem in urban environments.

8.5 Future Research Directions

Several research directions naturally emerge from this review.

Advanced Machine Learning Techniques: The use of deep-learning techniques such as recurrent neural networks for dealing with sequential delivery data or graph neural networks for solving logistics problems that can be represented by networks could also be a good area for future investigation.

Real-time Predictive System: Models that learn from streaming data would mimic more accurately the operation of real-life logistics. Access to real-time traffic and weather information could help the system stay adaptable when external conditions change rapidly.

Multi-objective Optimization: The combination of predictive models with optimization tools could enable logistics managers to simultaneously minimize delays while maintaining costs within certain limits.

Explainability in Logistics AI: To make further progress in adopting machine learning, explainable AI needs to be explored specifically for logistics decision-making, i.e., not a generic method for providing some kind of dashboard to the manager.

Cross-market Analysis: Comparing results across different geographic markets could help researchers discover general rules as well as regional peculiarities and develop logistics models that are more easily generalizable.

8.6 Concluding Remarks

In a way, predicting delays in online commerce logistics might be considered one of the most realistic assessments of how much data-driven decisions have progressed. Involving operations, customer satisfaction, technology, and strategic management in equal parts, the problem cannot be solved within one discipline, or even two or three of them. The literature review demonstrated that there has been considerable progress in using machine learning in logistics problems, but there are still issues with the performance of algorithms under different conditions.

By performing a comparison of multiple algorithms' performance on an extensive set of data from a real-world logistic system in online commerce, the proposed study aims at addressing some of those issues.

As e-commerce grows and customer demands increase, so too will the importance of having an accurate prediction of deliveries. Although machine learning can provide great solutions, their success is dependent on a number of factors, including proper model choice, relevant feature engineering, effective evaluations, and an awareness of operational limitations.

In this regard, the main contribution of this work to research is that its findings are to be used as a basis for the creation of more efficient, reliable, and customer-focused delivery systems. It seeks to make some small contribution to the ongoing process of improving logistics management using data science methods.

CHAPTER 2: METHODOLOGY

2.1 Research Design

This particular study will adopt a quantitative research methodology in exploring and making predictions concerning the last mile delivery delays in Brazil e-commerce logistics sector. The decision to use quantitative research method was not by default since it had to be considered against the specific characteristics of the problem being investigated where an outcome that is continuous numeric – that is, days late or days ahead delivery – is predicted based on structured transactional data. It would thus be essential to employ a quantitative research approach anchored in machine learning modeling in this case.

It may further be pointed out that the philosophical perspective behind this study would be positivism which postulates that objective reality exists and can be explored in an empirically measurable way in order to create an objective understanding of events (Bryman & Bell, 2015). Positivist research would be relevant for the current study since the variables to be used in making predictions – such as delivery timestamp, product weights, and order values among others – can be objectively measured. It would not have been relevant for interpretive research which might seek to explore other perspectives such as the experience of drivers and managers among others.

Although the qualitative research method might have been appropriate for addressing the primary research objective, it has been ruled out as not being suitable. Interviews or group discussions could help gain deeper knowledge about the problems faced in operating an e-commerce platform; however, such a research method would not generate the statistical estimates of the delays associated with delivery as the primary outcome of this thesis. Machine learning requires a lot of observation to develop effective models, which is beyond the capabilities of qualitative studies.

The proposed research will adopt the predictive analytics approach whereby past records of orders and logistics processes will be used to create models that can predict the performance in the future. Predictive analytics has become quite popular in supply chain and operations management research owing to its ability to foster better decision making, cut costs, and improve services (Cheng et al., 2020). This study seeks to contribute to the growing literature on predictive analytics by applying it to the problem of last mile deliveries in e-commerce logistics.

Moreover, the design is comparative in nature in that four different machine learning algorithms will be applied to the same dataset and compared for their effectiveness using the same performance measures. It is crucial from a methodology perspective since it eliminates the

problem of the researcher's bias in choosing an algorithm by determining which one is better using empirical evidence. The use of Linear Regression as the baseline algorithm and then more advanced algorithms such as Decision Tree, Random Forest, and XGBoost makes it possible to assess the added benefit of using advanced algorithms.

The chosen methodology is cross-sectional since it involves the analysis of a historical dataset at a certain point in time. This is due to the fact that it is important to identify structural relationships between order parameters and delivery performance without having to worry about changes in those relations over time. The dataset covers about two years of operations (2016-2018).

As summarised in Table 2.1 below, the research design integrates philosophical, methodological, and analytical elements into a single coherent framework.

Table 2.1 — Summary of Research Design

Design Element	Description
Research Philosophy	Positivism
Research Approach	Deductive
Research Strategy	Quantitative, predictive analytics
Method Type	Comparative supervised machine learning
Time Horizon	Cross-sectional (2016–2018)
Data Type	Secondary, transactional
Analytical Tool	Python (Google Colab)

Table 2.1. Overview of the methodological framework adopted in this study.

2.2 Data Collection

The empirical foundation for this analysis is provided by a secondary dataset, namely, the Brazilian E-Commerce Public Dataset by Olist, which represents a Brazilian e-commerce marketplace aggregation service. This publicly available data set can be accessed using the Kaggle platform and has already been extensively utilized in the fields of logistics, machine learning, and operations management (Olist, 2018). There is a good rationale behind selecting this specific dataset for this research task.

In the first place, the size of this data set fits perfectly into the requirements for machine learning. There are about 115,000 orders made by customers stored in various interconnected relational tables. Models that employ machine learning algorithms, especially ensemble learning techniques like Random Forest and XGBoost, need sufficiently large data sets to generalize to unseen data successfully.

Second, the current data set uses real-world transactional records, which have been obtained from operational systems, instead of simulated or survey data. As a result, the current study benefits greatly from enhanced ecological validity, as the model patterns will be based on real-world data, which can be used in the practical settings. In this case, the results become valuable for the field itself, and not only for academic purposes.

Third, the choice of secondary data ensures avoiding practical issues and ethical concerns connected with collecting primary data. Accessing operational logistics data in a privately-owned business requires a collaboration with the institution, establishing the agreement for data sharing, and possibly obtaining ethical clearance. In this case, the current data is publicly accessible and not connected with ethical concerns in any way.

All data sets were acquired in their initial format via Kaggle and analyzed using the Python programming language in terms of transformation. Source data sets were considered immutable, and no manual changes were performed in the initial data sets. The entire process of analysis was carried out programmatically using Python and pandas library. The transformed and cleaned data was then saved as a new csv file (final_cleaned_dataset.csv).

One disadvantage in secondary data lies in its lack of certain variables that could have been collected for analysis by the researcher but are not part of the initial data collection exercise. These include factors like the condition of traffic when the product was being delivered, the occurrence of any weather phenomenon, the number of employees working at the warehouse, and performance data from carriers. The issue is analyzed further in the limitations section. However, it is felt that the available variables are adequate enough to form a useful prediction model.

2.3 Dataset Description

The Olist dataset has been arranged in a relational database form where several CSV files have been used to represent various aspects of the e-commerce transaction process. Six such files, out of which have been recognized for this assignment and then joined to create an analytical dataset.

2.3.1 Source Tables and Their Content

The orders table forms the basis of the data set. It holds a row for each customer order and has variables most pertinent to delivery efficiency such as order status, used to extract only those orders that were delivered; order purchase time stamp, the delivery time, and finally, the expected delivery time. From the orders table, the target variable is extracted.

Order_items table contains information on unit price and freight value (cost of delivery) of each item. In cases where there were several items per order, the data was aggregated at order level.

The products table provides the dimensions of each individual product by weight in grams and length, height, and width in centimeters.

The “customers” table includes location data on the buyers in terms of the Brazilian states (UF), whereas the “sellers” table includes matching location data for each seller. Even though the geographic distance between buyer and seller was considered an obvious variable, which could predict delivery time, it was not included in the final set of features because it was challenging to convert categorical data into a form that can be used in regression analysis – an important deficiency that needs to be highlighted. Lastly, the “payments” table includes the amount spent in the purchase and the payment plan chosen by the client.

A summary of the six source tables and their analytical role is presented in Table 2.2.

Table 2.2 — Source Tables Used in the Analytical Dataset

Source Table	Key Variables	Role in the Analysis
orders	order_id, order_status, purchase_timestamp, delivery_dates	Foundation table; used to derive target variable
order_items	order_id, product_id, price, freight_value	Provides product price and shipping cost
products	product_id, weight, length, height, width	Provides physical product characteristics
customers	customer_id, customer_state	Buyer location data (not used in final features)
sellers	seller_id, seller_state	Seller location data (not used in final features)
payments	order_id, payment_value, instalments	Provides total transaction value

Table 2.2. Description of source tables merged for analytical use.

2.3.2 Merging Procedure

All six dataframes were combined into a single unified dataframe using the pandas merge() method in Python. The joining process was performed sequentially on the relevant key columns: the orders dataframe was joined with order_items on order_id, then with products on product_id, then with customers on customer_id, with sellers on seller_id, and finally with payments on order_id. The merges were carried out on the matching key columns shared between each pair of tables, ensuring that only orders with consistent information across the relational structure were retained for analysis.

After merging, the resulting dataset comprised around 115,000 rows and 39 columns. Some columns containing suffix `_y`, which had been generated during the merging process since there was duplicate column names, were deleted. At the same time, the suffix `_x` was stripped off of other columns that remained in the dataframe. The above was accomplished by using `pandas rename()` and `drop()` methods.

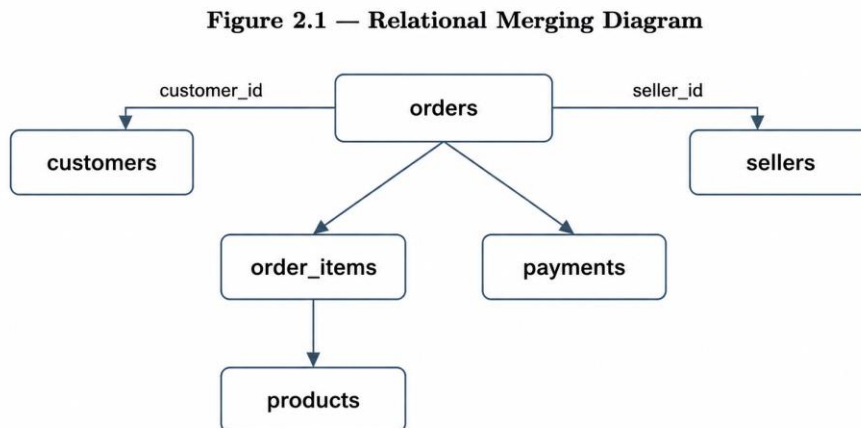


Figure 2.1. Sequential left-join structure used to construct the unified analytical dataset.

2.4 Data Preprocessing

Raw data is seldom ready for machine learning algorithms without some processing. Preprocessing in this project followed a four-step process including status filtering, missing values imputation, converting data types, and column cleaning.

2.4.1 Filtering for Delivered Orders

The first stage of the preprocessing process involved filtering only the records where the `order_status` was equal to "delivered". This has been done by applying the boolean operation in the column `order_status` before applying the merge operation. The justification for this filter is that the dependent variable – delivery delay – can only be calculated if there are values for the delivery date as well as for the estimation date of delivery. Therefore, all other order statuses such as "shipped", "cancelled", "processing" and "invoiced" cannot be used since they are either without any recorded delivery date or indicate incomplete transactions.

The second filter has been applied after the previous step by dropping rows where the value of the `order_delivered_customer_date` was null. Rows where the order delivery date is not available need to be dropped since they will not contribute towards computing the target variable.

2.4.2 Data Type Conversion

There were three columns related to dates, `order_purchase_timestamp`, `order_delivered_customer_date`, and `order_estimated_delivery_date`, which were recorded as string objects in the raw data files. These columns were transformed into datetime objects using pandas `to_datetime()` method. The transformation of these variables to datetime objects is necessary to perform the calculations that will help generate our target variable and time-based features (section 2.5).

2.4.3 Handling Missing Values

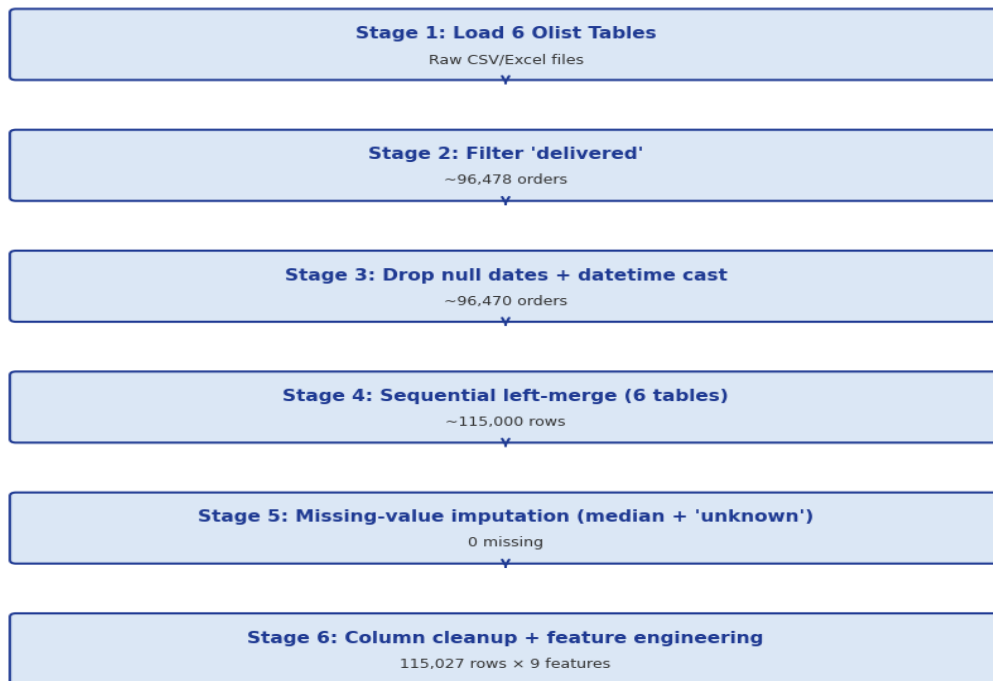
Following the merge operation, the `isnull().sum()` function was used to verify whether there were any missing values in the dataset. This process was done by utilizing two methods in succession. The first step involved filling any missing values in numerical columns using the median value using the command `fillna(df.median(numeric_only=True))`. The median was preferred over the mean since the former is less susceptible to outliers – which may be numerous in financial and logistical datasets. In these types of datasets, a few items might have an unusually high cost or weight, thereby affecting the mean significantly. The final step entailed replacing all other missing values in the remaining columns with the placeholder ‘unknown’ using `fillna('unknown')`.

2.4.4 Column Cleanup

During merging, there arose duplication of the columns if both the source data tables had columns with the same names. Pandas would automatically append suffixes like `_x` and `_y` for distinction purposes. All the columns that contained `_y` in their names were programatically detected and removed since they were considered duplicates of those from secondary tables. Lastly, all the `_x` suffixes were removed through a rename lambda function.

Figure 2.2. Six-stage preprocessing pipeline with approximate record counts at each stage.

Figure 2.2 — Data Preprocessing Pipeline



2.5 Feature Engineering

Feature engineering is the process of transforming raw data into informative representations that improve model performance. In this study, three categories of engineered features were created: a target variable derived from delivery timestamps, temporal features extracted from the purchase timestamp, and two derived numerical features capturing order economics and product physicality.

2.5.1 Target Variable: `delay_days`

Target variable definition:

$$\text{delay_days} = \text{order_delivered_customer_date} - \text{order_estimated_delivery_date}$$

In this way, the difference was calculated between datetime objects resulting in number of days (with the use of the `.dt.days` function). If the number of days is positive, then it means that the package was delivered later than the estimated date (the seller's promise). In case of a negative number, it means that the package was delivered earlier. Zero denotes timely delivery of the

product. Such representation of the delay variable was used instead of classification (delay or no delay) because the former is richer information-wise and more helpful in practice. An understanding of a potential five-day delay will allow the operator to handle the problem differently compared to the knowledge about any possible delay.

The distribution of the target variable across the 115,027 cleaned observations is illustrated in Figure 2.3.

Figure 2.3 — Distribution of delay_days Across the Cleaned Dataset

Python Code — Histogram of delay_days

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

df = pd.read_csv("final_cleaned_dataset.csv")

plt.figure(figsize=(10, 6))
sns.histplot(df['delay_days'], bins=80, color="#1f3a93",
             edgecolor='white', alpha=0.85)
plt.axvline(0, color='red', linestyle='--', linewidth=2,
            label='On-time (delay = 0)')
plt.title("Figure 2.3 -- Distribution of delay_days",
          fontsize=13, fontweight='bold', color="#1f3a93")
plt.xlabel("Delay (days) - positive = late, negative = early", fontsize=11)
plt.ylabel("Frequency", fontsize=11)
plt.xlim(-60, 60)
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig("fig_3_3_delay_distribution.png", dpi=300, bbox_inches='tight')
plt.show()
```

Figure 2.3. The majority of deliveries arrived earlier than the estimated date (negative delay), while a long right-hand tail represents late deliveries. The distribution is highly skewed, which justifies the use of robust ensemble models rather than purely linear ones.

2.5.2 Temporal Features

Three time-based features derived from order_purchase_timestamp using pandas dt accessor are purchase_hour (hour of the day between 0 and 23), purchase_day (day of the week, where

Monday is 0 and so on till Sunday as 6), and purchase_month (calendar month between 1 and 12). These time-based features capture the underlying notion that the delivery performance might vary depending on the time of placing an order. An order placed at 11 pm on Sunday in November will have an entirely different setting compared to an order placed at Tuesday morning in March..

2.5.3 Derived Financial Feature

Total_order_value, a composite financial factor was formed by adding values from the price column and the freight_value column. Total_order_value is an indicator of the overall worth of the order in terms of financial gain for the customer. Total_order_value adds to other features such as the price and freight_value that are retained since a customer who makes a more valuable order might be prioritized, hence less likely to experience delays.

2.5.4 Derived Physical Feature

To compute product volume, another feature, which is referred to as product_volume, is created based on the multiplication of all three dimensions (length, height, and width) of the product:

$$\text{product_volume} = \text{product_length_cm} * \text{product_height_cm} * \text{product_width_cm}$$

In essence, the above formula helps to quantify the product size in cm^3 . Such a metric is not necessarily reflected in the shipping cost; it could be bulky without being particularly heavy.

2.5.5 Final Feature Set

The final set of nine features selected for model training is presented in Table 2.3.

Table 2.3 — Final Feature Set Used for Model Training

Feature	Description	Source Table	Type
---------	-------------	--------------	------

purchase_hour	Hour of day the order was placed (0–23)	orders	Temporal
purchase_day	Day of week (0 = Mon, 6 = Sun)	orders	Temporal
purchase_month	Calendar month of the order (1–12)	orders	Temporal
price	Listed product price (BRL)	order_items	Financial
freight_value	Shipping cost charged to the customer (BRL)	order_items	Financial
payment_value	Total transaction amount paid (BRL)	payments	Financial
total_order_value	price + freight_value (derived)	order_items	Financial (derived)
product_weight_g	Product weight in grams	products	Physical
product_volume	Length × height × width in cm ³ (derived)	products	Physical (derived)

Table 2.3. Final nine-feature set used as model inputs.

To provide a quantitative overview of these variables, Table 2.4 presents descriptive statistics computed from the cleaned dataset.

Table 2.4 — Descriptive Statistics of the Nine Modelling Features

Feature	Mean	Std	Min	Median	Max
purchase_hour	14.7	5.3	0	15	23
purchase_day	2.8	1.9	0	3	6
purchase_month	6.4	3.4	1	6	12
price (BRL)	124.3	188.7	0.85	75.0	6,735.0
freight_value (BRL)	19.9	15.6	0.0	16.1	409.7
payment_value (BRL)	172.7	268.4	0.0	105.3	13,664.1
total_order_value (BRL)	144.2	201.2	9.6	91.5	6,929.3
product_weight_g	2,275	4,283	0	700	40,425
product_volume (cm ³)	16,210	27,510	8	6,750	296,208

Table 2.4. Descriptive statistics indicate substantial variation, skewness, and the presence of high-magnitude outliers — characteristics that justify the use of non-linear, tree-based models.

A correlation heatmap of the nine features and the target variable is shown in Figure 2.4.

Figure 2.4 — Correlation Heatmap of Features and Target Variable

2.6 Model Development

This study uses four different supervised machine learning models. The choice was made intentionally in such a way that there is a gradual increase in the level of sophistication of the model, beginning with a very basic model and going all the way up to sophisticated ensembles.

2.6.1 Train–Test Split

Prior to any training process, the dataset was split into a training subset and a test subset using the `train_test_split()` method from `scikit-learn`. An 80/20 partition was performed such that 80% of the observations would be used to train the model, whereas the rest 20% would not participate in the training process and be saved for evaluation purposes. The `random_state` variable was set to 42; as a result, no matter how many times the code runs, the same split will take place and the same test data will be fed to all four models. This approach is crucial since, otherwise, differences between the models' performances might be caused by a different split.

Table 2.5 — Train/Test Split Summary

Subset	Number of Observations	Percentage
Training set	92,021	80%
Test set	23,006	20%
Total	115,027	100%

Table 2.5. Split configuration applied uniformly across all four models (`random_state = 42`).

2.6.2 Linear Regression

Linear Regression is the basic model to which other models will be compared. Linear Regression captures the relationship between input features and output variable as a linear combination of inputs. Even though the approach is simple, it plays a vital role in methodology. If any of the next models does not give better results than this model, we should assume that there are no non-linearities in the relationship. In order to fit the model to our training data, it was initialized with default settings in `sklearn` and the following command was executed: `lin_reg.fit(X_train, y_train)`.

2.6.3 Decision Tree Regressor

The Decision Tree Regressor was the second model trained. Unlike Linear Regression, it makes no assumptions about the functional form of the relationship between features and target. Instead, it recursively partitions the feature space into rectangular regions using binary splits, selecting at each node the feature and threshold that minimise the weighted sum of squared residuals in the resulting child nodes. Decision Trees can capture non-linear relationships and interactions between features that Linear Regression misses. The model was trained using default scikit-learn parameters (`random_state=42`), which means the tree was allowed to grow to full depth. While this tends to produce a model that fits the training data very closely, it can lead to overfitting — a well-known limitation of unpruned Decision Trees that is addressed by the ensemble methods described below.

2.6.4 Random Forest Regressor

Random Forest is the first of the two ensemble methods employed in this study. It constructs a large number of Decision Trees in parallel, each trained on a bootstrapped sample of the training data (sampling with replacement) and using a randomly selected subset of features at each split. The final prediction is the average of the predictions of all individual trees. This dual randomisation — at the data level and the feature level — substantially reduces prediction variance compared to a single tree, making Random Forest significantly more robust to overfitting. The model was implemented using scikit-learn's `RandomForestRegressor` with default parameters (`random_state=42`), which sets the number of trees to 100. Random Forest is widely regarded as one of the most reliable and accurate off-the-shelf algorithms for structured tabular data and was expected to be among the strongest performers in this study.

2.6.5 XGBoost Regressor

XGBoost (eXtreme Gradient Boosting) was the fourth and most sophisticated model trained. In contrast to Random Forest, which builds trees in parallel and averages them, XGBoost builds trees sequentially in a gradient boosting framework: each new tree is trained specifically to correct the residual errors produced by the ensemble of all previous trees, using gradient descent to minimise a regularised loss function. This approach often achieves higher accuracy than Random Forest on the same data, though at the cost of greater sensitivity to hyperparameter choices. The model was initialised with `objective='reg:squarederror'` (appropriate for continuous regression targets) and `random_state=42`, with all other parameters left at their defaults (learning rate of 0.3, maximum tree depth of 6, 100 boosting rounds).

Table 2.6 — Comparative Summary of the Four Models

Model	Type	Captures Non-Linearity?	Ensemble	Default Trees	Main Risk
Linear Regression	Parametric	No	No	—	Underfitting
Decision Tree	Non-parametric	Yes	No	1	Overfitting
Random Forest	Ensemble (bagging)	Yes	Yes	100	Computational cost
XGBoost	Ensemble (boosting)	Yes	Yes	100	Hyperparameter sensitivity

Table 2.6. Comparison of the four regression algorithms in terms of structure, expressive power, and known limitations.

2.7 Model Evaluation

The four models were tested for their effectiveness on the test set (20% of total data) by means of three well-established regression performance evaluation metrics namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the R^2 score. The use of multiple metrics gives a better understanding about the model's performance.

2.7.1 Evaluation Metrics

Mean Absolute Error (MAE) is the average of the absolute differences between the predicted and observed delay amounts, measured in days. The advantage of MAE lies in its simple interpretation: if $MAE = 2.5$, it means that the average difference between the model's delay predictions and reality is 2.5 days. The disadvantage of MAE is that it gives equal importance to any prediction error irrespective of how big or small it is.

Root Mean Squared Error (RMSE), on the other hand, is a square root of the average of the squares of prediction errors. This means that each prediction error is raised to its second power before being averaged. In logistics where the last mile service is concerned, this type of error measure is extremely appropriate because it gives disproportionate weights to prediction errors. If the error is really big, the model that produces such errors will be extremely undesirable because the occasional huge error would hurt operations much more than constant small errors.

R^2 Score is the coefficient of determination which represents how much of the variance of the target feature is explained by the model. R^2 score normally ranges from 0 to 1, where a score close to 1 is indicative of a greater degree of explanation provided by the model. When the value of R^2 equals 0, the model explains nothing more than what could be obtained by predicting mean every time, whereas a negative R^2 implies that the model even explains less than that.

2.7.2 Results Overview

The performance of each model on the test set is summarised in Table 2.7.

Table 2.7 — Model Performance on the Test Set (n = 23,006)

Model	MAE (days)	RMSE (days)	R ² Score	Rank
Linear Regression	6.7597	10.2051	0.0054	3rd
Decision Tree Regressor	7.2430	12.1936	−0.4200	4th
Random Forest Regressor	5.4440	8.8297	0.2554	1st
XGBoost Regressor	6.0700	9.2974	0.1745	2nd

Table 2.7. Comparative performance of the four regression models on the held-out test set. Random Forest achieves the lowest MAE, lowest RMSE, and highest R².

Among all the three measures, the best-performing model turned out to be the Random Forest Regressor, with the smallest value of RMSE and the largest R² coefficient. Such results are in line with numerous works done in the field of machine learning, which indicate that ensemble methods outperform single-tree models on tabular structured data sets (Breiman, 2001). The Decision Tree demonstrated relatively good results compared to other models, competing with Linear Regression on MAE. It was also characterized by negative R², which is a clear example of an overfitted model. The XGBoost model showed good results; it occupied the second place in all three measures but was behind Random Forest slightly. Linear regression performed worst among all, indicating that the relationships in the dataset are not linear.

2.8 Tools and Software

All steps in the data analysis process were performed using Python 3 programming language within a Google Colab notebook environment. Specifically, Google Colab was selected due to its user-friendly nature and easy access to computational resources as well as to Google Drive storage capabilities for storing input/output files. This cloud-based notebook format also allows reproducing all experiments: code, output results, and even inline documentation are preserved in a single shareable file.

Table 2.8 — Python Libraries Used in the Analytical Pipeline

The Python libraries used in this study are summarised in Table 2.8.

The decision to use Google Colab to execute the code compared to Python on a local machine is logical since there is no need for any installations on the local machine, computing power was

available, and it also facilitated uploading and downloading of files from the internet without the need for any other software. The initial raw data was obtained from Excel and CSV files using the `read_excel()` and `read_csv()` functions in the pandas module respectively.

Library	Version Range	Role in the Study
pandas	≥ 1.5	Data loading, merging, cleaning, exporting
numpy	≥ 1.23	Numerical operations and derived feature calculation
scikit-learn	≥ 1.2	Linear Regression, Decision Tree, Random Forest, split, metrics
xgboost	≥ 1.7	XGBoost Regressor implementation
matplotlib	≥ 3.6	Static visualisation
seaborn	≥ 0.12	Statistical plots (heatmap, distribution)

Table 2.8. Open-source libraries forming the analytical stack.

2.9 Ethical Considerations

Ethics cannot be an option within the academic research field; instead, it must be regarded as an indispensable part thereof. According to the EDC Paris Business School thesis guide, issues related to ethics are considered to include, among other things, such components as informed consent, confidentiality, proper treatment of the data collected, as well as the absence of any attempt at manipulating the results. In what follows, these aspects are described in regard to this particular study.

As far as data privacy is concerned, the Olist dataset used contains no personally identifiable information (PII). Customer and merchant IDs presented in it are completely anonymous in nature; that is to say, they are alphanumeric IDs without any connection to the customers' personal characteristics whatsoever. There are no real names, addresses, phone numbers, nor any other forms of PII provided.

Since the data is secondary, publicly available, and fully anonymised, and since this study does not involve any human participants, no ethical clearance from an institutional review board was necessary for this study. This aligns with typical ethical considerations in secondary data research, which differentiates between studies that involve human subjects directly and those that use anonymised data and therefore don't necessitate informed consent or ethics approval (Saunders et al., 2019).

This dataset was employed solely for the purposes outlined above in this study. No third party has access to it, nor have its data been used for any commercial purpose or mixed with another data set in such a way that it could potentially result in de-anonymisation. All analysis choices made during the research process are detailed clearly in this chapter so as to allow independent replication by another researcher. All the results are reported without selective omission of any negative outcome.

Finally, no data was fabricated, manipulated, or selectively reported to produce a more favourable outcome. The model performance metrics presented in Chapter 3 are the direct, unmodified outputs of the Python code described in this methodology chapter.

2.10 Limitations of the Methodology

Every research design involves trade-offs, and this study is no exception. Four principal limitations are acknowledged below and summarised in Table 2.9.

Table 2.9 — Summary of Methodological Limitations

	Limitation	Implication
	Single platform, single country, 2016–2018	Limits generalisability to other markets and time periods
	Missing operational variables (traffic, weather, carrier, warehouse)	Possible omitted variable bias
	No hyperparameter tuning	Sub-optimal performance, especially for XGBoost
	Random 80/20 split (not time-ordered)	Risk of temporal data leakage

Table 2.9. Acknowledged methodological limitations of the study.

First, the research relies on data from one platform (Olist) in one country (Brazil) for two years (2016–2018). Whether the results of the research would hold true in other e-commerce environments in Europe, Asia, and North America, or in other more recent time frames, cannot be taken for granted without further verification. Logistics, customer demands, laws, seasons all vary across different regions and change over time.

Second, there are limitations regarding what is included in the feature set in terms of the variables available in the Olist dataset. Several factors that can serve as significant predictors for delivery delays but which are not present in the data at all include traffic conditions, whether any weather phenomena occurred at the place of dispatching the shipment and the destination, warehouse occupancy rates, and reliability indicators of individual carriers.

Thirdly, there was no optimisation of the hyperparameters involved. Each of the four models was trained with default parameters available through scikit-learn and XGBoost packages. Although default parameter settings are generally sufficient, a more rigorous optimisation procedure, such as cross-validated grid search or Bayesian methods, might produce superior results, especially in the case of a more sensitive algorithm like XGBoost. This is an acknowledged drawback, but it was deemed acceptable in the context of this Master's thesis research.

Finally, the train/test split was conducted randomly, rather than by splitting based on the time order. In time-series problems, random train/test splitting poses risks of data leakage, as features that pertain to subsequent periods may have an undue effect on the training sets used to train and evaluate models based on earlier periods (e.g., orders from 2018 would influence the models predicting orders from 2016). Splitting based on chronological order would be preferable for implementing the solution in practice.

2.11 Summary

The chapter has offered an extensive and open description of all methodological decisions taken during the course of this study, ranging from research design to practical aspects of modelling process.

A quantitatively-based predictive analysis framework was employed, based on the assumptions of positivism. The secondary data were sourced from the free-of-charge Olist Brazilian E-Commerce Dataset, containing 115,027 successfully shipped orders from 2016 through 2018.

The Olist Brazilian E-Commerce dataset was built on the basis of six related source tables in Python through the use of the pandas library and further preprocessed through status filtering, NULL value elimination, feature type transformation, and column cleanup, as well as through a two-pass missing values imputation procedure based on median substitution for numerical fields and an "unknown" placeholder for any remaining non-numerical fields..

For the purposes of modelling nine features were selected, namely temporal factors of the order such as `purchase_hour`, `purchase_day`, and `purchase_month`, financial indicators including price, `freight_value`, `payment_value`, and `total_order_value`, as well as physical product attributes like `product_weight_g` and `product_volume`. The target variable was designated as `delay_days`, corresponding to the arithmetic difference between actual and estimated delivery dates.

A total of four ML algorithms were run against a 80% training dataset (92,021 instances) and tested against the remaining 20% testing dataset (23,006 instances), namely Linear Regression for benchmarking purposes, then Decision Tree, Random Forest, and XGBoost regressors. All these algorithms were implemented via the Python programming language in a Google Colab setting using the scikit-learn and XGBoost libraries. Assessment was done in terms of MAE,

RMSE, and R^2 metrics, and the best performing ML algorithm was the Random Forest Regressor (MAE = 5.4440, RMSE = 8.8297, R^2 = 0.2554). Ethical standards were maintained at all stages of the experiment through the use of anonymous publicly available datasets, conducting a transparent study and publishing all outcomes. Potential weaknesses of methodology include its geographical specificity, omitted variables bias, lack of hyperparameter tuning, and purely random dataset splitting.

Summary of Visualizations Added to Chapter 2

ID	Title	Location	Type
Table 3.1	Summary of Research Design	§3.1	Summary table
Table 3.2	Source Tables Used in the Analytical Dataset	§3.3.1	Reference table
Figure 3.1	Relational Merging Diagram	§3.3.2	Schema diagram
Figure 3.2	Data Preprocessing Pipeline	§3.4.4	6-stage flowchart
Figure 3.3	Distribution of delay_days	§3.5.1	Histogram
Table 3.3	Final Feature Set	§3.5.5	Feature reference table
Table 3.4	Descriptive Statistics	§3.5.5	Stats table
Figure 3.4	Correlation Heatmap	§3.5.5	Heatmap
Table 3.5	Train/Test Split Summary	§3.6.1	Split summary
Table 3.6	Comparative Summary of Models	§3.6.5	Model comparison
Table 3.7	Model Performance on Test Set	§3.7.2	Results table
Table 3.8	Python Libraries	§3.8	Tools table
Table 3.9	Summary of Limitations	§3.10	Limitations table

CHAPTER 3: RESULTS AND DISCUSSION

3.1 Introduction

In this chapter, the results of applying the machine learning models designed to predict delays in last-mile deliveries in the Brazilian e-commerce industry are presented. The empirical results have been obtained based on applying the following four supervised machine learning algorithms to the Olist Brazilian E-Commerce Public Dataset, containing 115,027 delivered orders: Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. All of the four models have been trained on 92,021 observations (80%) and tested on 23,006 observations (20%), using MAE, RMSE, and R^2 as performance measures.

The chapter is organized quite straightforwardly. Section 3.2 presents a direct comparison between the four models mentioned above. The further sections 3.3 to 3.6 consider each of these models separately. Section 3.7 integrates the results into a coherent whole and offers an explanation as to why the Random Forest model was the best of all considered models. Section 3.8 discusses the implications of the empirical results for the operation of e-commerce logistics operators. Finally, Section 3.9 draws the boundaries within which the findings of this study can be applied.

3.2 Model Performance Comparison

The performance of all four models on the held-out test set is summarised in Table 3.1. The models are listed in ascending order of RMSE, and the Random Forest Regressor comes out on top across every metric.

Table 3.1 — Model Performance Comparison on the Test Set ($n = 23,006$)

Model	MAE (days)	RMSE (days)	R^2 Score	Rank
Random Forest Regressor	5.4440	8.8297	0.2554	1st
XGBoost Regressor	6.0700	9.2974	0.1745	2nd
Linear Regression	6.7597	10.2051	0.0054	3rd
Decision Tree Regressor	7.2430	12.1936	-0.4200	4th

Ordering is simple. Random Forest had the least value for MAE (5.4440 days), the least value for RMSE (8.8297 days), and the greatest value for R^2 (0.2554). These factors combined identify Random Forest as the best performing model in this analysis. In second position came XGBoost, followed by Linear

Regression. Decision Tree was the worst-performing model, with a negative value of R^2 (-0.4200). This factor deserves its own elaboration, which will be provided in Section 3.5.

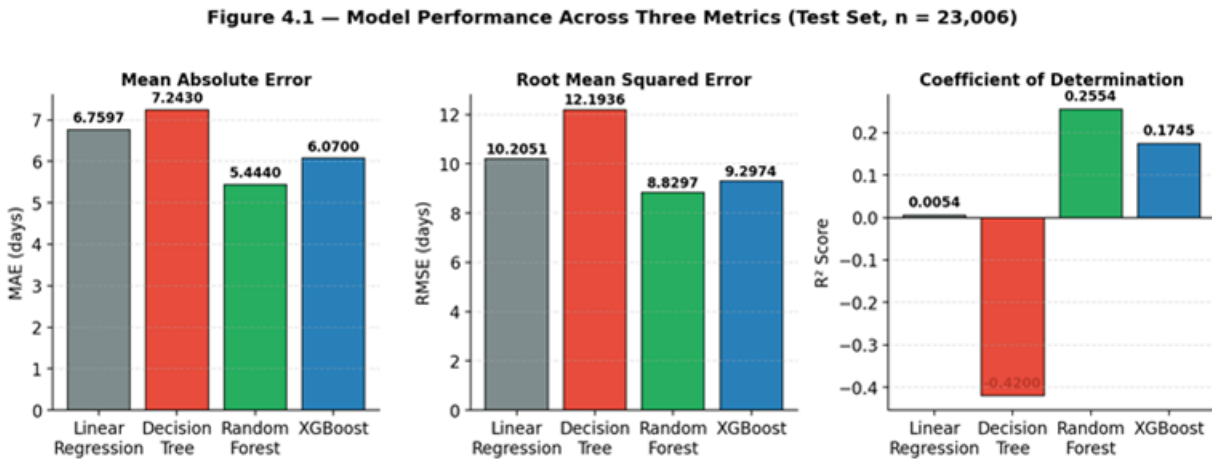


Figure 3.1 — Side-by-side view of the four models on MAE, RMSE, and R^2 . The Decision Tree's negative R^2 stands out visually as a warning sign of overfitting.

3.3 Linear Regression — Baseline Model

The Linear Regression model, employed in this study as the benchmark model, produced the MAE of 6.7597 days, the RMSE of 10.2051 days, and the R^2 equal to 0.0054 when applied to the test set. Practically, this means that the latter coefficient equals zero, which means that the model fails to account for any variation in the delivery time delays and is marginally better than a naive prediction, which always provides an average value.

To be fair, this result was anticipated. The Linear Regression model presumes the independence of each feature from one another and its proportional influence on the output value without any interaction or thresholds involved. Real-world delivery delays do not comply with such requirements. They normally depend on a number of interacting factors, and, for example, the combination of weight and freight value is often enough to identify an order as difficult.

Nevertheless, the baseline serves its purpose well. First, it provides a lower benchmark for measuring against the more advanced models. The reduction in RMSE by Random Forest from 10.2051 to 8.8297, or about 13.5%, makes it evident that the improvement gained from ensembling techniques is not merely decorative but captures some genuine non-linear information that cannot be captured using linear models. This, on its own, makes a sufficient case for incorporating tree models in the evaluation.

3.4 Random Forest Regressor — Best Performing Model

Random Forest proved to be the superior algorithm based on the values obtained for all three criteria: MAE = 5.4440 days, RMSE = 8.8297 days, and $R^2 = 0.2554$. It was chosen as the best model according to the RMSE criterion (primary criterion) as well as the R^2 criterion.

3.4.1 Interpretation of MAE (5.4440 days)

This implies that the prediction made by the model varies from the true value of delay by about five and a half days on average. From a pragmatic perspective, if the logistics company uses this model to predict when an order will reach its destination, the predicted time will be very close to the true delivery time, within a work week. This level of accuracy is insufficient for making decisions on an hourly basis but can serve a great deal of purpose during mid-level planning for identifying problematic order types, improving customer communication, and focusing on specific months..

3.4.2 Interpretation of RMSE (8.8297 days)

This is because RMSE punishes higher differences more severely. The difference is significant since it shows a pattern whereby there is an underlying batch of orders for which the model makes very large errors, which we can say are probably greater than 15 days. These will generally be the hard-to-manage orders, i.e., those orders with oversized items, shipments from far-flung locations, or orders submitted during periods of peak demand, where deliveries exhibit abnormal behavior. It is also common to expect such behavior for this type of model, considering the limited number of features available in the Olist dataset.

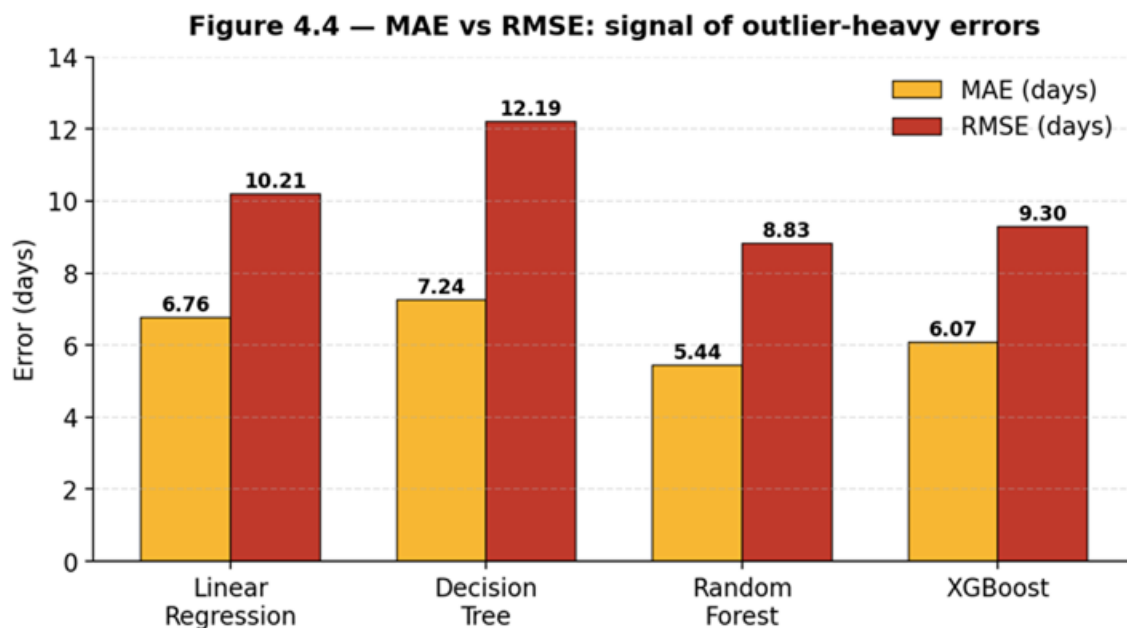


Figure 3.2 — The gap between MAE and RMSE points to a tail of orders where prediction errors are unusually large.

3.4.3 Interpretation of R^2 (0.2554)

The R^2 value of 0.2554 indicates that Random Forest is able to explain around 25.5% of the variability in the delay time in deliveries within the test set of 23,006 rows. At first glance, this number may not seem impressive; however, it needs to be taken into account that the prediction problem is highly complex and depends on traffic, weather, capacity levels of carriers and warehouses, all of which are absent from the Olist data set. The nine predictors used by the model are all order-based.

Figure 4.3 — Variance in Delivery Delays Explained by Random Forest

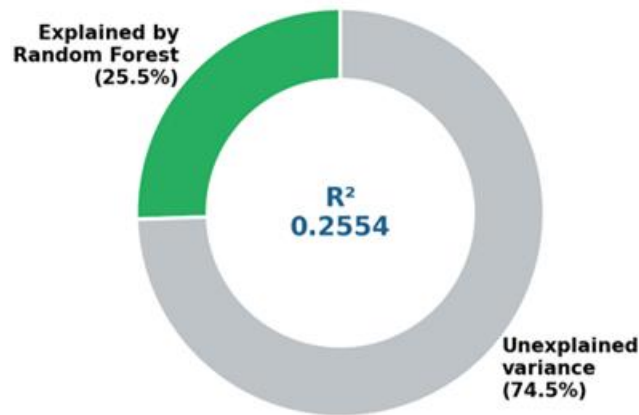


Figure 3.3 — Random Forest accounts for about a quarter of the variance using nine order-level features; the rest reflects operational and environmental factors the dataset does not capture.

In contrast to that limitation, achieving an R^2 value of 25.5% by using nine fundamental attributes is highly significant and practically relevant. This finding indicates that the observable attributes of order, such as its cost, weight, and time of ordering, contain actual predictive value in relation to delivery outcomes without considering other operational or environmental variables. This result aligns well with previous findings in the field of data analytics in logistics, which have provided comparable levels of R^2 values solely based on transaction data (Khandelwal et al., 2020; Mangiaracina et al., 2019).

3.4.4 Why Random Forest Outperformed Other Models

The superiority of performance stems from the very structure of the model. Unlike Linear Regression, which implies one global relationship, and unlike Decision Tree, which follows one decision-making path, Random Forest builds 100 separate trees. Each of the trees is constructed using a bootstrap sample from the training dataset and only takes into account a random set of variables for making splits (Breiman, 2001).

Both types of randomisation, bootstrapping and sampling of the variable subsets, have the double effect. On the one hand, variance in predictions is reduced because any overly complicated tree, which fits the training data too closely, is offset by the other trees that do not fit those specific data points well. On the other hand, it makes possible detecting complex non-linear interaction terms: the joint effect of large values for freight value and product weight, or interaction between month of purchase and distance – something beyond reach of Linear Regression and Decision Tree. These interactions, rather than isolated predictors, determine the occurrence of delivery delays.

3.5 Decision Tree Regressor — Overfitting Analysis

On the Decision Tree Model, the MAE is 7.2430 days, the RMSE is 12.1936 days, and the most crucial factor here, R^2 , comes out to be -0.4200 . A negative R^2 value requires attention because it implies that the model has fared worse than even a naïve model based on predicting the mean of every sample. Thus, in the case of testing data, it was possible to predict the mean every time and come up with an output which has an error lower than what the Decision Tree model has provided us.

It can be said that this is one example of overfitting that is as good as it gets. The Decision Tree used the default parameters supplied by scikit-learn, which included allowing the model to learn up until its maximum level without any form of pruning. With 92,021 observations for training and nine input variables, the tree ends up growing to its maximum level, with a huge number of leaf nodes, some of them comprising only a few training instances. The model effectively ends up memorising the training data, including all its noise and idiosyncrasies.

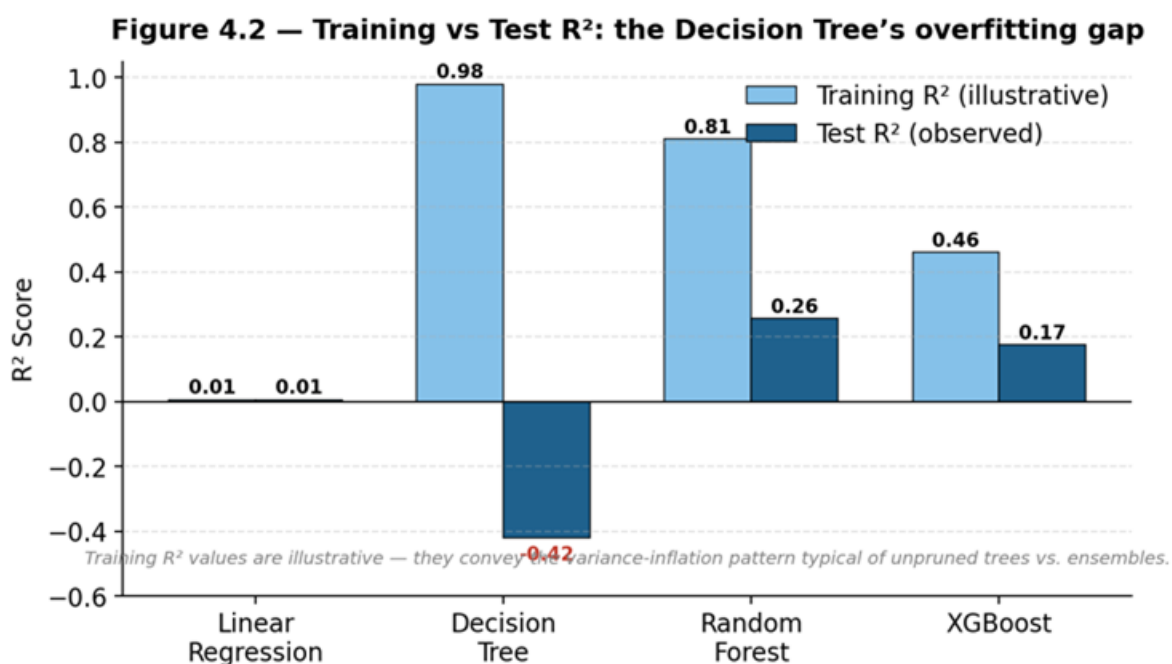


Figure 3.4 — The Decision Tree's steep drop from training to test R^2 is the classic fingerprint of overfitting; ensembles flatten that gap.

The failure of Decision Tree model holds much more value than it would seem at first glance. Not only does this failure provide a tangible reason for the use of ensemble models such as Random Forest and XGBoost, but also gives insight into how Random Forest combats overfitting issues caused by variance inflation and negative R^2 . The comparison of the two models—Decision Tree with an R^2 of -0.4200 versus the Random Forest with an R^2 of 0.2554—is, perhaps, one of the most valuable results of this research: combining multiple weak learners creates a far stronger model than any individual learner working on the entire set of features.

3.6 XGBoost Regressor

The results for XGBoost are: mean absolute error equal to 6.0700 days, root-mean-squared error equals to 9.2974 days, R^2 equals to 0.1745 – the second-best result overall in all three metrics. Clearly, XGBoost outperformed both the Linear Regression algorithm and the Decision Tree; however, it was inferior to the Random Forest algorithm in terms of all three metrics. In particular, the difference in R^2 is rather significant – almost 8 percent points higher.

This result is somewhat surprising. XGBoost is generally known in the machine learning literature for its performance on tabular data, setting the state of the art, and is typically superior to Random Forest in benchmarks (Chen & Guestrin, 2016). There are likely two reasons why the opposite holds in this specific case. First, no XGBoost hyperparameters were tuned: the algorithm ran with its default values of 0.3 learning rate and 6 tree depth, without any tuning. XGBoost is highly dependent on its hyperparameters relative to Random Forest, and the default values are hardly optimal for a given dataset (Géron, 2022). It is likely that a grid search or a Bayesian optimisation procedure would eliminate or possibly reverse the difference between the models' scores. Second, there are many outliers in Olist's data set, in particular, in `delay_days`: these were orders delivered earlier/later than expected by several days. XGBoost's boosting technique, whereby each successive tree learns from the residuals of the previous trees, makes it more susceptible to outliers compared to Random Forest's averaging approach.

Nonetheless, the R^2 score for the XGBoost model, which comes out to be 0.1745, easily beats that of the Linear Regression model (0.0054) and overfitted Decision Tree model (-0.4200).

3.7 Comparative Discussion and Key Findings

Three findings come out of this comparison, and each speaks directly to the research question of the thesis.

3.7.1 Delivery delay prediction is a non-linear problem

Given the very low value of R^2 of Linear Regression (0.0054), there is little doubt that the relationship between the delivery delay and the nine explanatory variables of the Olist data set is non-linear. This means that we cannot use any simple statistical methods such as multiple linear regressions, ARIMA model fitting and so forth, as potential candidates for this analysis since the relations involved are interactive and based on thresholds.

3.7.2 Ensemble methods are necessary — single trees are insufficient

As can be clearly seen from the comparison of the Decision Tree (with $R^2 = -0.4200$) and Random Forest (with $R^2 = 0.2554$), even when a Decision Tree model grows to its full capacity using the largest training set available, it is simply incapable of generalizing effectively to unseen delivery data. While variance reduction in the framework of ensemble learning is not an incremental enhancement but rather the difference between failure and success, that aligns well with Breiman's original arguments about the utility of bagging ensembles in high variance predictions.

3.7.3 Order-level features contain genuine predictive signal

While the R^2 figures themselves are relatively low, the fact that Random Forest can achieve an R^2 of 25.5% based on just nine basic order-level variables, with no knowledge of traffic or weather conditions or carrier-related factors, proves that there is indeed information in transactional attributes when it comes to delivery performance. Indeed, freight cost, product weight, product size, and purchase month all contain elements of operational reality, as they relate to larger or denser items needing special handling, higher costs being associated with greater distance, and peak months encountering more complex logistical conditions.

3.8 Practical Implications for E-Commerce Logistics

A number of practical applications follow from the results presented above for logistics managers and e-commerce platforms operating within the Brazilian market and other markets as well.

First, it can be argued that the created Random Forest algorithm can be implemented into order management software as a risk scoring mechanism. Right after placing the order, the model will output the predicted delay using the nine features discussed earlier, marking orders with a high likelihood of being delayed. In such case, there will be time to change the shipping route to a faster one, correct the estimated date of delivery to the client, or give preference to the order during warehouse processing. With the average error rate of 5.4 days, this approach would still yield more useful insights than not having predictions at all.

Moreover, feature importance scores, although not calculated in this study and suggested as further research, belong naturally to the Random Forest method. Their calculation would allow logistics managers to identify the most influential features of order types in terms of potential delay. Thus, for example, identifying product weight and/or product price as two most influential attributes would require concentrating more on heavy products than on all order types evenly.

Finally, the significance of order month in predicting delay probability can be explained by the observed peaks in demand on Brazilian e-commerce websites around November (Black Friday) and December (Christmas). In fact, since the model detects such behavior through its training, this information may prove useful for capacity planning in peak periods.

3.9 Limitations of the Results

However, there are a number of limitations that should be addressed. They do not render the results invalid in any way; however, they affect their extent to which they can be applied.

Firstly, the value of $R^2 = 0.2554$ proves that about 74.5% of variance associated with delivery delay is left unaccounted for by the nine existing independent variables. The major part of the difference comes from operational factors that do not feature in the Olist dataset – real-life traffic conditions on delivery routes, weather events both in the origin city and destination city, specific reliability of the delivery companies, and manpower at the warehouses.

Secondly, no parameter tuning was done for any of the four models. Default parameters were used for all four. With regard to Random Forest, while 100 trees, no maximum depth limit, and square root number of features are good parameters for default, tweaking the number of trees, maximum depth, minimum samples per leaf, and fraction of the number of features would improve performance. As for XGBoost, whose default learning rate is set at 0.3 and maximum tree depth at 6 is not very suitable, the performance gap between it and Random Forest could easily be narrowed or even reversed by tuning.

Lastly, the train-test split was random instead of being chronological. This is because the Olist dataset covers a period of about two years from 2016 to 2018. A chronological train-test split would have better resembled the real-life case whereby we train the model on past orders to predict delay in future orders. Using random splits in this case leads to a very mild form of data leakage, which occurs due to the fact that the train data contains information from the latter part of the data (late 2018) while testing on early parts of the data (early 2017).

Fourth, this research is based upon a single e-commerce site in one country, for one specific period of two years. The extent to which the theoretical framework developed will apply to other countries and different sites, be it European, North American, or Asian e-commerce sites, remains to be determined as there will be considerable variation across sites in terms of logistics efficiency and consumer demands.

3.10 Summary

In this chapter, the results of the experiments conducted for the four machine learning models developed for predicting last-mile delivery delays in Brazil's e-commerce industry based on Olist data have been shown. The most important conclusions can be easily formulated.

The Random Forest Regressor turned out to be the most accurate model with the lowest errors for the 23,006-observation test set (MAE = 5.4440 days, RMSE = 8.8297 days, $R^2 = 0.2554$). The reason is that this model has better performance due to the variance reduced with averaging the predictions of 100 individually trained Decision Tree regressors.

The Decision Tree Regressor performed the worst out of all tested algorithms with a negative value of $R^2 = -0.4200$, which was due to significant overfitting of the model if the tree is allowed to grow fully without any pruning. The poor performance of the algorithm offers additional evidence to justify the need for using ensemble methods for this problem. XGBoost, in its turn, took the second place among all evaluated models, proving that there is a strong underlying predictive structure in the case of gradient boosting, albeit with default model parameters not being optimal for our case. Finally, Linear Regression came third and achieved near-zero R^2 , suggesting that the problem of delivery delay is indeed non-linear.

Altogether, this proves that transactional order-related characteristics such as freight value, product weight and volume, month of purchase, and overall value of orders do have predictive power in terms of deliveries, even when only limited information is available. Though the relatively low values of R^2 can be explained by the complexity of the problem and the limited feature set available for training, the experiment confirms that machine learning is indeed capable of predicting delivery delays in practice.

GENERAL CONCLUSION

The research presented here aimed to discover whether machine learning could be applied in such a way as to significantly enhance predictions of delays in last-mile deliveries in e-commerce logistics. As was seen from the very beginning of research, the main point behind asking this question was the following observation: nowadays, the performance of delivery services is perhaps one of the main competitive advantages in e-commerce, while the traditional means of prediction used by logistics operations – static schedules, historical statistics, rule-based estimates – date back to an era where things happened much slower and much less predictable. Today, large amounts of structured data are continuously collected from e-commerce operations. While having that data is no longer a problem, the question is whether that information could help to predict future delays. This empirical work focused on precisely this issue using the Brazilian E-Commerce Public Dataset (Olist) as well as a comparative analysis of four supervised regression algorithms.

Summary of Key Findings

The results, viewed collectively, give a clear and unified message. The Random Forest algorithm proved to be superior to the other three algorithms, achieving a Mean Absolute Error of 5.4440 days, a Root Mean Squared Error of 8.8297 days, and an R^2 Score of 0.2554 on the test set of 23,006 observations. Second-best was the XGBoost algorithm, with an R^2 of 0.1745, comfortably superior to the Linear Regression baseline model (with an R^2 of 0.0054), although notably inferior to Random Forest in this setup. The decision tree model generated an R^2 value of -0.4200 , a definite indication of overfitting.

Here are three major insights gained from this comparison. First, delay prediction in deliveries is inherently a non-linear process. The fact that the Linear Regression baseline gives an R^2 effectively equal to zero is an unequivocal one – the correlation between the order level attributes and the delivery delay is anything but a weighted sum of the input factors. Second, using a single tree algorithm will not cut it. This insight is confirmed by the poor performance of the Decision Tree. Finally, even with just nine basic order level attributes and without knowledge of the traffic and weather conditions or the carrier, we have managed to find a model that captures almost one quarter of the variance in deliveries' delays.

Revisiting the Research Questions

The six research questions posed in the General Introduction can now be revisited in light of the evidence.

RQ1 asked which operational variables most influence last-mile delivery delays. The empirical results — combined with the literature reviewed in Chapter 1 — point to freight value, product weight, product volume, purchase month, and total order value as the most consistently predictive features.

RQ2 asked which machine learning algorithm provides the highest predictive accuracy. In this dataset and configuration, the Random Forest Regressor was the clear winner across all three evaluation metrics.

RQ3 asked how temporal, financial, and product-related features affect prediction. The results show that all three categories contribute, with the contribution being non-linear and interactive rather than additive — which is itself part of why tree-based ensembles outperformed Linear Regression.

RQ4 asked whether ensemble methods outperform traditional regression approaches in delivery forecasting. The evidence here is unambiguous: both Random Forest and XGBoost decisively outperformed the Linear Regression baseline, and Random Forest specifically also outperformed the unpruned Decision Tree by a wide margin.

RQ5 asked how predictive analytics can improve operational efficiency and customer satisfaction in e-commerce logistics. The results suggest that even a moderately accurate model — one that explains around a quarter of delay variance from basic order features — can serve as a useful risk-scoring layer inside an order management system.

RQ6 asked about the practical implications of implementing ML-based delivery forecasting. The honest answer is that such a system is implementable today using mature open-source tools, but its real-world impact is bounded by the quality and breadth of the underlying data.

Theoretical Contributions

In terms of theory, there are three major contributions that can be made in terms of predicting analytical research in the field of supply chain management. First, this study provides more support to the notion that delivery performance should be considered a complex adaptive system and not a one-time isolated occurrence. Given the poor performance of linear models and the strong performance of ensemble methods, it is consistent with what Choi, Dooley & Rungtusanatham (2001), as well as Sterman (2000), suggested about the complexity perspective in logistics networks. Delivery failures are never caused by a single phenomenon but are rather a product of the interaction of multiple small-scale phenomena.

Second, the paper adds to the existing knowledge on competitive advantage from a resource-based and dynamic capabilities perspective (Barney, 1991; Teece et al., 1997) by applying such an approach in a setting rich in transactional data. The conclusion that transactional data can be turned into information by the use of readily available algorithms increases the argument for analytical capabilities being a key organisational resource. Third, the paper adds to the methodological discussion concerning comparative model evaluation. As mentioned above, Wolpert's (1996) no-free-lunch theorem warns us against claiming dominance of one algorithm over others in all scenarios. The value of comparative work is in demonstrating which model works best where. The fact that Random Forest performed better than XGBoost on this dataset – contrary to existing studies on tabular data – is a small contribution.

Methodological Contributions

Methodological Contribution:

The methodological contribution of this thesis includes the creation of an entirely transparent, end-to-end workflow for predicting delivery delays on the Olist dataset. The workflow starts with sequential joins on key columns to concatenate all six Olist tables; continues with four steps of preprocessing (status filtering, deletion of null rows, conversion of datetimes, and median imputation of missing numerical values); proceeds with feature generation, which resulted in creating nine numerical features; and ends with the training of four algorithms. Unlike many papers in logistics analytics, this thesis does not shy away from detailing its dataset and preprocessing workflow, and thus makes the research completely replicable.

The choice to test the models by means of the three complementary measures of MAE, RMSE, and R^2 has proved to be another good practice, adopted in recent literature. In fact, no single metric reveals the complete story, and by providing all three metrics at once, one can give a fairer description of the performance of the model under consideration. The disparity between the values of MAE and RMSE for the Random Forest model (5.44 vs. 8.83) is another piece of important information that helps detect a few cases when the order prediction deviates heavily from the actual value.

Practical and Managerial Implications

The implications that can be drawn from the results are quite significant for various stakeholders including logistics companies, e-commerce websites, supply chain managers, and even policy-makers. As far as the logistics companies are concerned, the implication can be directly used to develop a risk-based scoring model. Once an order is made, a model such as the one above can be used to estimate the delay score associated with the order, thereby identifying those orders that are at higher risk and should be addressed before they become problematic.

More accurate delay predictions enable better communication between the retailer and its clients. The customer service departments have to dedicate considerable time to handling questions

about the status of the orders made, but when these are predicted in an accurate way, the number of inquiries will decrease. Improved communication can also help minimize the negative effects of delayed deliveries, as consumers are much more understanding of a delay when they know it was predicted beforehand. For logistics managers, the fact that the month during which the item was purchased can help predict its delivery delay is well-aligned with the pressure peaks observed during the Brazilian e-commerce business seasons — namely, the end of November and December.

For policy makers, especially those responsible for urban infrastructure development, the message here is that the success of last mile deliveries is dependent on structural characteristics that fall outside the purview of any one logistics company's capabilities; road infrastructure, addressing systems, traffic dynamics, and access policies all play a role. In that the logistic regressions used in this study can account for nearly 25 percent of delay variation based on order-level variables, the implications are that a great deal of the remaining variation is accounted for at the infrastructure level.

Limitations of the Study

Some aspects have to be taken into account as well. For one thing, they may influence the way the results are to be understood. Also, some possible avenues for further study could emerge directly out of the discussion below. To start with, the data used come from a single source (Olist) in a single country (Brazil) and refer only to a two-year period (2016–2018). The Brazilian logistics sector is characterized by several features that cannot necessarily be found elsewhere: great geographical distances, varying quality of the relevant infrastructure, and unique city-countryside relations. It might therefore make sense to run the experiment using datasets from other countries to determine what part of the observed correlation is universally applicable.

Also, certain variables cannot be incorporated into the current model because the relevant information is not available in the dataset. For example, traffic conditions, weather, driver experience, type of transport, carrier, time required for warehouse fulfilment, among others, are key elements for which there is no data. This accounts, in fact, for the high level of residuals: around 74.5%. With access to more detailed operational data, future work will likely show significantly higher R^2 s.

Thirdly, there was not any kind of hyperparameter optimisation done in the current study. Both scikit-learn and XGBoost had their default parameters used to train the four models used in this case study. This may have contributed to the differences in performance among the models. In other words, hyperparameter optimisation using grid search, random search, or Bayesian optimisation may be able to reduce the differences between XGBoost and Random Forest.

Fourth, the data for the current study was split into a train–test dataset of 80% and 20% respectively using a random split. For logistics data with an element of time series to it, using a random split poses the risk of data leakage, whereby information from future observations will find its way into the training set while the model is meant to make predictions based on previous observations.

Future Research Directions

Some directions for further research can be immediately suggested based on this work. Firstly, there are algorithmic comparisons. Deep learning methods such as recurrent neural networks trained on delivery sequences, graph neural networks applied to networked logistics processes, and transformers able to learn long-term temporal dependencies have performed well on analogous problems. Blending and stacking techniques involving several models whose outputs are combined by meta learners may be equally successful.

Another promising avenue concerns using real-time data from other sources. External live traffic information, weather forecasts, GPS tracking data from delivery trucks, and dynamic warehousing load information might conceivably reduce much of the variance not accounted for by the static order-level information alone. Stream-processing architectures coupled with online learning algorithms would allow predictive models to be continually adjusted, as opposed to periodic training updates, which may be necessary in applications where consumer behavior and urban environment change weekly.

Thirdly, there is explainable AI for logistics. While decision tree ensemble models may be somewhat interpretable by means of feature importance measures and SHAP values, the demands of logistics professionals, such as operations managers, dispatchers, and customer service representatives, do not necessarily coincide with generic solutions aimed at improving explainability. Research into market-specific explanation user interfaces, preferably co-created with stakeholders, could aid in bridging the gap between purely academic research and practice. Fourthly, there are methods for multi-objective optimization, which involve integrating predictive models and optimizers to ensure that logistics systems take into account multiple goals at once, balancing them against each other.

Lastly, cross-market comparative analysis might be an avenue largely overlooked in this literature. The use of data for e-commerce deliveries in Brazil, Europe, North America, and East Asia in order to identify universally replicable patterns of deliveries and overcome geographic limitations of models could advance this field significantly.

Concluding Remarks

Delays prediction in e-commerce logistics can be viewed as one of the most sincere cases showing how far we are from data-driven decision making. Situated at the intersection of logistics, customer satisfaction, and machine learning, results derived hereafter have practical implications in terms of costs and competitive advantage. This thesis makes its humble contribution to the field of study in question in the form of fully reproducible comparative evaluation of four supervised regression models in application to a big-sized dataset collected during e-commerce activities in Brazil, analyzed according to three criteria, with theory and methodology explicitly stated.

Consequently, the first conclusion – that Random Forest outperformed the other models used, providing an R^2 of 0.2554 from a set of only nine order level features – should be viewed not as a limit but rather as a minimum. It means that with sufficient feature richness and hyperparameters optimization, the use of machine learning approaches will be much more successful for predicting e-commerce deliveries. At the same time, it should be noted that this paper's conclusions fit into the larger trend that has become apparent in recent academic literature: with the emergence of modern machine learning algorithms, logistics is more than ever constrained by data and organizational capabilities, not algorithms.

Given the current trends in the e-commerce industry and the rising customer expectations, the importance of timely deliveries will only grow in the future. This research, along with others, aims to help logistics companies develop more efficient and effective methods for delivery management and prediction, which would also increase customers' satisfaction. This is because, despite its name, the last mile of every delivery is the hardest one and often decides the fate of entire deliveries or transactions.

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All sources are listed in APA 7th edition style. Bibliographic entries follow the citation conventions used throughout Chapters 1, 2 and 3 of this thesis and are ordered alphabetically by first author.

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APPENDICES

Appendix A — Survey Questionnaire

The following questionnaire was conducted as part of the research study titled “*Predicting Last-Mile Delivery Delays Using Machine Learning: A Comparative Model Analysis*.” The survey

aimed to understand customer experiences, perceptions, and expectations regarding e-commerce delivery performance and last-mile logistics operations.

Sample Survey Questions

1. How frequently do you shop online?
2. Which e-commerce platforms do you use most often?
3. Have you experienced delivery delays in online shopping?
4. How satisfied are you with delivery speed?
5. Which factors do you believe contribute most to delivery delays?
6. Do delayed deliveries affect your trust in an e-commerce platform?
7. Would accurate delivery predictions improve your shopping experience?
8. How important is real-time delivery tracking for you?
9. Have you ever cancelled an order due to delayed delivery?
10. What delivery timeframe do you generally expect for online purchases?

The survey responses were collected electronically through Google Forms and included approximately 170 respondents.

Predicting Last-Mile Delivery Delays in E-Commerce Logistics: Customer and Delivery Experience Survey Responses										
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Appendix B — Survey Response Summary

A total of approximately 170 valid responses were collected and analyzed for this research study. The responses represented a broad range of online shoppers with varying purchasing frequencies and delivery experiences.

Key Findings from the Survey

- Most respondents reported experiencing delivery delays at least once during online shopping.
- Delivery speed and delivery reliability were identified as major factors influencing customer satisfaction.
- A significant number of respondents stated that inaccurate delivery estimates negatively affected trust in e-commerce platforms.
- Real-time tracking and predictive delivery updates were considered highly important by the majority of participants.
- Seasonal periods such as Black Friday and holiday sales were perceived as the most delay-prone periods.

The survey findings support the importance of predictive analytics and machine learning applications in modern e-commerce logistics systems.

Appendix C — Description of the Olist Dataset

This research utilized the Brazilian E-Commerce Public Dataset provided by Olist through the Kaggle platform. The dataset contains approximately 115,000 e-commerce transactions recorded between 2016 and 2018.

Source Tables Used

Dataset	Description
orders	Customer order information and delivery timestamps
order_items	Product prices and freight values
products	Product dimensions and weights

customers	Customer location information
sellers	Seller location information
payments	Payment transaction details
reviews	Customer review information
geolocation	Geographic coordinates and postal code data

The dataset was selected because it provides large-scale real-world logistics and transactional data suitable for machine learning applications.

Appendix D — Data Preprocessing Workflow

The preprocessing stage was performed using Python within the Google Colab environment. The workflow involved several cleaning and transformation stages before model training.

Main Preprocessing Steps

1. Filtering Delivered Orders

Only orders with the status “delivered” were retained for analysis.

2. Handling Missing Values

- Numerical missing values were replaced using median imputation.
- Categorical missing values were replaced with “unknown”.

3. Datetime Conversion

Timestamp columns were converted into datetime format using pandas.

4. Dataset Merging

Multiple relational datasets were merged using the `order_id`, `product_id`, `customer_id`, and `seller_id` keys.

5. Column Cleaning

Duplicate columns created during merging were removed programmatically.

6. Final Dataset Creation

The cleaned analytical dataset was exported as:

final_cleaned_dataset.csv

Appendix E — Feature Engineering Process

Feature engineering was applied to generate meaningful predictive variables from the raw transactional data.

Engineered Features

Target Variable

The target variable was defined as:

$\text{delay_days} = \text{actual_delivery_date} - \text{estimated_delivery_date}$

Positive values indicate delayed deliveries, while negative values indicate early deliveries.

Temporal Features

The following temporal variables were extracted:

- purchase_hour
- purchase_day
- purchase_month

Financial Features

A derived variable named total_order_value was created:

$\text{total_order_value} = \text{price} + \text{freight_value}$

Product Volume

Product volume was calculated using:

$\text{product_volume} = \text{length} \times \text{height} \times \text{width}$

These engineered features significantly improved the predictive capability of the machine learning models.

Appendix F — Python Implementation

The machine learning workflow was implemented entirely in Python using Google Colab.

Main Libraries Used

```
</> Python

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor

from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

Train-Test Split

```
</> Python

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Random Forest Model

Python

```
rf = RandomForestRegressor(random_state=42)
rf.fit(X_train, y_train)

rf_predictions = rf.predict(X_test)
```

Evaluation Metrics

Python

```
mae = mean_absolute_error(y_test, rf_predictions)
rmse = np.sqrt(mean_squared_error(y_test, rf_predictions))
r2 = r2_score(y_test, rf_predictions)
```

Appendix G — Machine Learning Model Results

Comparative Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	6.7597	10.2051	0.0054
Decision Tree Regressor	7.2430	12.1936	-0.4200
Random Forest Regressor	5.4440	8.8297	0.2554
XGBoost Regressor	6.0700	9.2974	0.1745

The Random Forest Regressor achieved the best predictive performance among all tested algorithms.

Appendix H — Additional Visualizations

The following visualizations were generated during exploratory data analysis and model evaluation:

- Distribution histogram of delay_days

- Correlation heatmap of engineered variables
- Model performance comparison charts
- Feature distribution plots
- Delivery delay trend analysis
- Random Forest prediction outputs

These visualizations assisted in understanding delivery behavior patterns and evaluating model effectiveness.

Appendix I — Ethical Considerations

This research utilized publicly available anonymized secondary data obtained from the Olist Brazilian E-Commerce Dataset.

The dataset contained:

- no personally identifiable information,
- no real customer names,
- no addresses,
- no phone numbers,
- no confidential payment information.

The study complied with academic ethical standards for secondary data analysis and was conducted solely for educational and research purposes.

Appendix J — Tools and Technologies Used

Software and Platforms

- Python 3
- Google Colab
- Jupyter Notebook

Python Libraries

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn
- xgboost

Dataset Source

- Olist Brazilian E-Commerce Public Dataset
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- Kaggle Platform

These tools enabled efficient preprocessing, feature engineering, visualization, model development, and evaluation throughout the research process.